VJ VIZZIES

USER GUIDE & VIDEO COMPANION

The Big Easy Guide

Everything you need to run the show: install, the control window, layers and looks, FX on the beat, scenes, the APC mini, cameras and the grandMA3.



FOOK'N OATS ENTERPRISES
SUITE 0.3.15 · NOVEMBER 2026

TESTED ON THE LIVE RIG

Contents

24 chapters. New here? Read 01, 06, 14 first.

01 Start here

02 Install, update and first launch

03 How the rig connects

04 Starting and stopping the player

05 Network, output and mapping

06 The control window

07 Layers, clips and the clip library

08 Generators, looks, shapes and overlays

09 FX, strobe and the beat clock

10 Audio react

11 Blend modes and busking stacks

12 Scenes: save and recall a look

13 Recording

14 The APC mini

15 Phone page and iPad remote

16 Keyboard shortcuts

17 Cameras: the hub and OATS CAMS

18 Driving it from grandMA3 and OSC

19 Safety: strobe cap, standby and test card

20 Security

21 Troubleshooting

22 Show-day checklist

23 Explainer video plan

24 Glossary

How this guide was made. Every step was tried on the running player before it was written down: the screenshots are the real control window, APC mini page and camera hub. Chapter colours follow the layer colours in the app: cyan, pink, amber, lime, violet.

01 Start here

VJ VIZZIES is a live visuals player for shows: you stack clips, fractals and generated looks on layers, add FX that lock to the beat, and send the result full screen to the LED wall or projector. You run it from a browser page, an APC mini, or the grandMA3.

What's in the suite

PART	WHAT IT DOES	WHERE YOU OPEN IT
VJ VIZZIES player	Renders the show at 1920x1080, 60 fps: a fractal background plus 4 layers	Runs in the background; the show picture goes out HDMI
Control window	Every control for the player, with live monitors	VJ VIZZIES.app, or a browser at `http://<this Mac>:8090`
APC mini page	A copy of the APC mini on screen; the real APC mini drives it	`http://127.0.0.1:8090/apc` in Chrome or Edge
OATS NDI HUB	Cameras as NDI sources, a multiview, PTZ control, tally	`http://<this Mac>:7080`
OATS CAMS	Menu-bar app that runs only the cameras	The house icon in the menu bar
VJ VIZZIES.app	Installer, updater, health dot in the menu bar, start / stop / restart	Applications folder

The 60-second tour

1. Open VJ VIZZIES. The menu-bar dot goes green when the player is running.
2. The wall shows the standby card (the logo) until you bring up a look.
3. In the control window, pick a clip or a look on layer L1 and push its opacity up.
4. Add an FX from the layer's FX slot, and set the tempo with Tap or Ableton Link.
5. Watch the monitor in the control window: that is the live output.
6. Save the look as a scene so one click brings it back.

If you only read one other chapter, read **The control window** and **The APC mini**.

02 Install, update and first launch

Installing takes one download, one drag and two questions. The example clip library comes inside the download.

What the Mac needs

- macOS 14 (Sonoma) or later, Apple Silicon or Intel (the 2020 Intel MacBook Pro works).
- The NDI runtime. NDI Tools installs it; if it's missing, the installer adds the bundled copy and asks for your password once.
- On Apple Silicon, Rosetta 2. The installer adds it if it's missing.
- A connection to the lighting network, ideally on one network port only.

Install from the .dmg

1. Download `VJ-VIZZIES-SUITE-<version>.dmg` from oats-apps-hub.vercel.app/vizzies and open it.
2. Drag **VJ VIZZIES** onto the **Applications** folder beside it, then open it from Applications.
3. First time on a Mac, macOS blocks it because the app isn't notarized yet. Go to System Settings > Privacy & Security, scroll down, click **Open Anyway**, then open it again. You only do this once per Mac.
4. The setup window asks two things:
   - **NDI machine name**: this Mac's name on the network. Every Mac needs its own name, or receivers only see one of them.
   - **Lighting interface**: the network port on the lighting network (en0, en1 ...).
5. Click **Install**, then click **Allow** when macOS asks for the camera, microphone and local network.

It installs two folders, `~/Applications/VJ VIZZIES` and `~/Applications/OATS NDI HUB`, starts everything, and opens itself at login.

Update

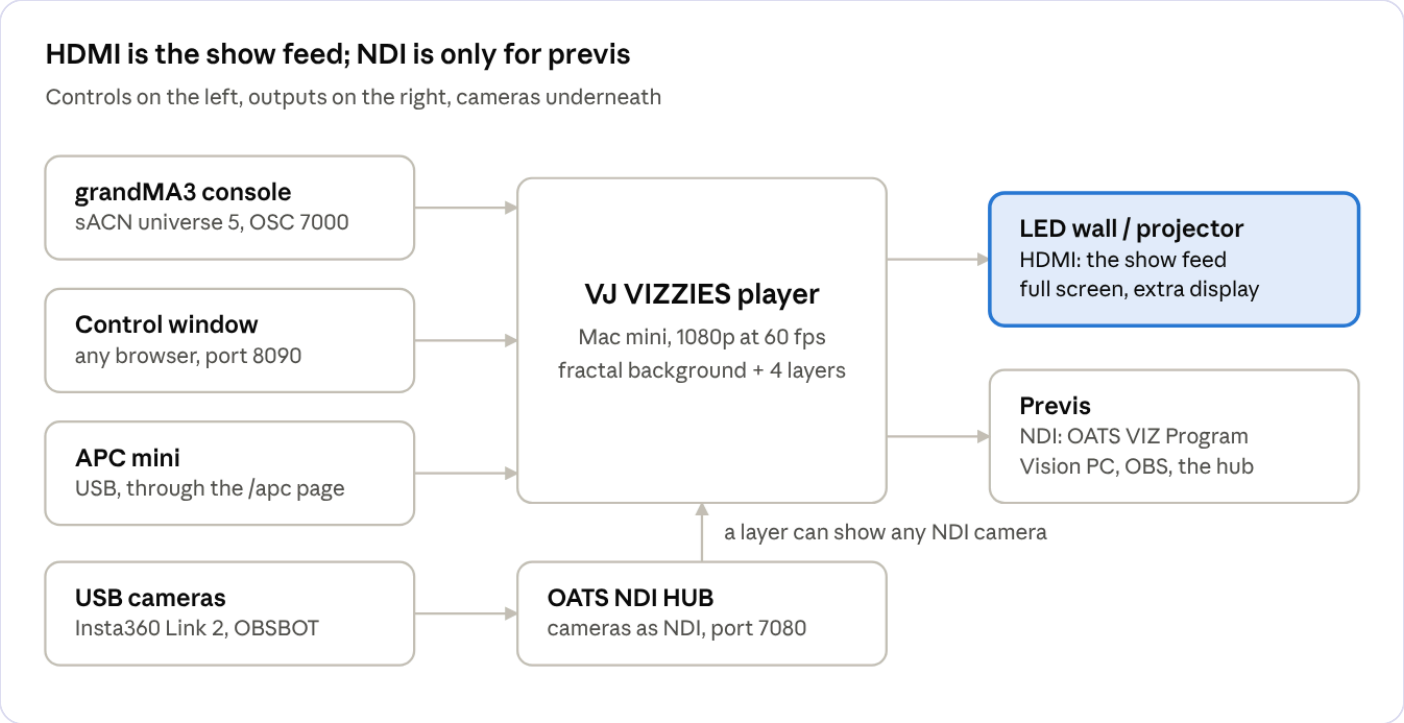
- Click **Check for Updates...** in the VJ VIZZIES app menu. It also checks once a day by itself.
- It downloads the new version, replaces itself and updates both folders. Your settings, presets, scenes, camera profiles and clips are kept.
- The player restarts during an update, so update between sets, never mid-show.

Cameras only? Use **OATS CAMS** (the house icon). It installs and runs only the camera hub, never the player, and updates itself.

Scripted install (for techs): unzip `VJ-VIZZIES-SUITE-<version>.zip` and double-click `INSTALL.command`. It asks the same two questions.

03 How the rig connects

The player sits in the middle: controls come in from the left, the show goes out HDMI on the right, and NDI copies the picture to previs.



The show rig: 3 controls, 2 outputs, cameras underneath

- **The show feed is HDMI.** The player runs full screen on an extra display: the LED processor or projector. That's what the audience sees.
- **NDI is for previs only.** "OATS VIZ Program" goes to Vision on the Windows PC, OBS or the hub multiview. Never send the show over NDI.
- **grandMA3** drives the player as a fixture on sACN universe 5 (import `gdtf/OATS@OATS VIZ.gdtf`) or with OSC on port 7000.
- **The control window and the APC mini** talk to the player over the network on port 8090. Any phone or laptop on the lighting network can open it.
- **Cameras** plug into the Mac by USB. The hub turns each into an NDI source, and any player layer can show one.
- Two Macs on one network need different NDI names and, if both listen, different sACN universes.

04 Starting and stopping the player

The player starts by itself when the Mac logs in, and the menu-bar dot is the easiest way to start, stop or restart it.

The menu-bar dot

(from VJ VIZZIES.app)

- Green: the player is running. Click the dot for **Start**, **Stop**, **Restart**, **Logs** and **Open at login**.
- If the player crashes or freezes, it restarts itself within about 4 seconds (the wall is black meanwhile). A stop you asked for keeps it stopped.

What the wall shows at start

- A cold start shows the **standby card**: the VJ VIZZIES logo on a slowly moving navy background. It holds the wall until you bring up a look.
- A restart skips the card but comes back with an empty layout, not your look: recall a scene to bring it back.
- The control window plays a short boot screen the first time it opens. Press any key to skip it.

From Terminal

(for techs)

COMMAND (RUN IN `~/APPLICATIONS/VJ VIZZIES/OATS-M0`)	WHAT IT DOES
`./m0.sh status`	Is it running, and how healthy
`./m0.sh stop`	Stops the player and keeps it stopped
`./m0.sh restart`	Restarts it without the standby card
`./m0.sh log`	Follows the player's log
`../autostart.sh now`	Starts everything the login item starts

Keep your look across a restart: `python3 tools/oats_state.py save` before, and `python3 tools/oats_state.py restore <file>` after. Scenes do the same job from the control window.

05 Network, output and mapping

The rig runs on its own lighting network, 192.168.1.x behind the Netgear router, and the show Mac sits on it by cable at 192.168.1.148. The player only takes control from that network and from the Mac itself.

The lighting network

(as set up in October)

MACHINE	ADDRESS	ROLE
Mac mini (show Mac)	192.168.1.148, wired	Player, control window, NDI out; its Wi-Fi joins the building network for internet only
PC "vectorworks-ma3"	192.168.1.141	grandMA3 onPC and Vision
DJ Mac "EDMOATMEAL"	192.168.1.15 wired	djay: tempo over Ableton Link, music over NDI
iPad and phone	on the Netgear's Wi-Fi	The remote pages
Netgear R7000	192.168.1.1	Router for the lighting network; travels with the rig

Rules that keep it working

- Never join the show Mac's Wi-Fi to the Netgear's own Wi-Fi. Two addresses on one network send NDI over Wi-Fi.
- The control window (port 8090), the previews (8091), OSC (7000) and sACN (5568) refuse anything outside the lighting network. A phone or iPad must be on the Netgear's Wi-Fi to reach them.
- NDI is pinned to the Mac's wired address. If that address ever changes, restart the player.
- The player's NDI source is **OBS-SERVER--2- (OATS VIZ Program)**: the machine name from setup, then the source name. Vision finds it by that name, so don't rename the Mac's NDI name before a show.

Output

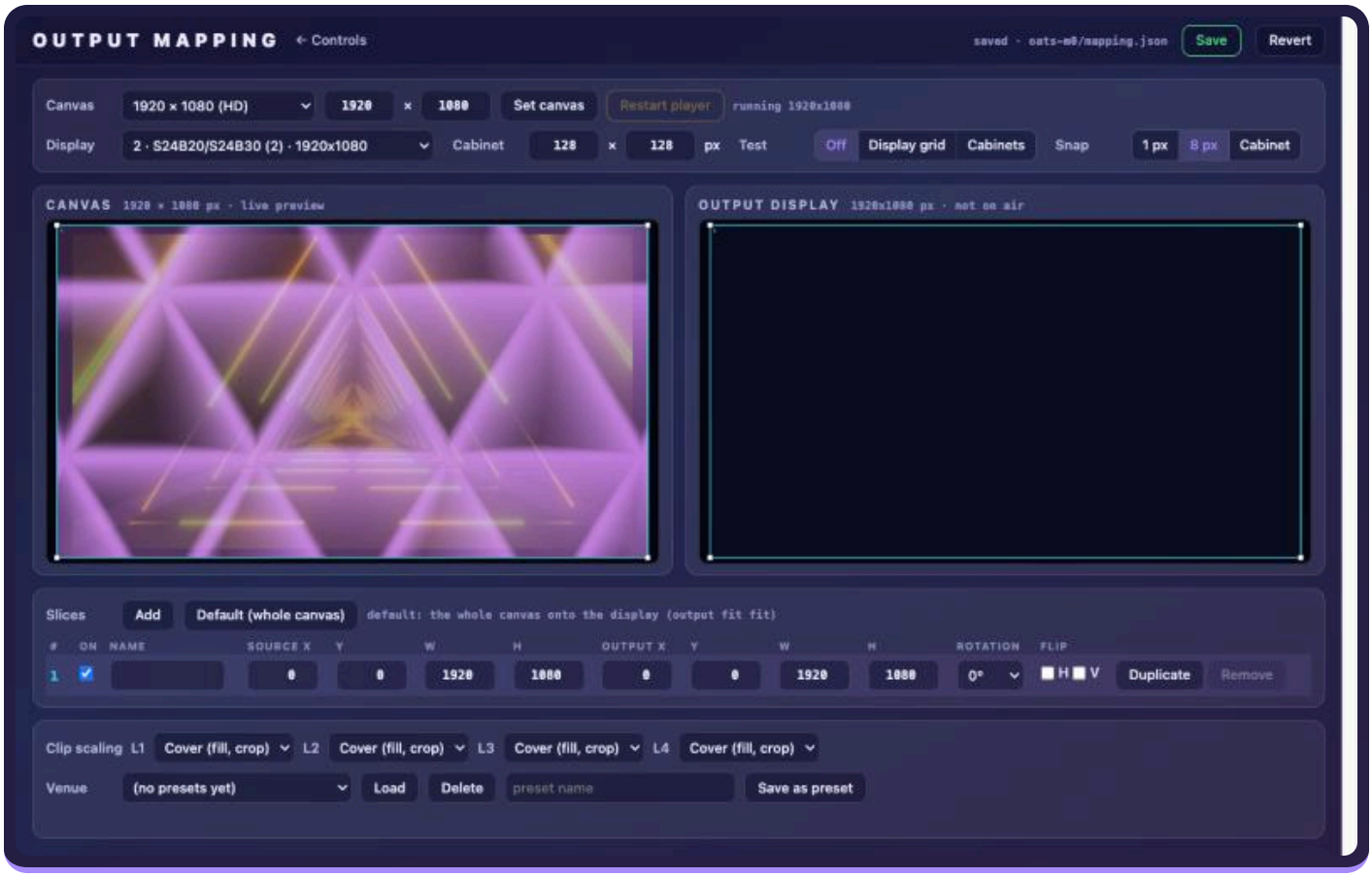
- The player renders 1920x1080 at 60 frames a second. When a deep fractal dive gets heavy, it lowers the fractal's detail for a moment instead of dropping frames.
- Pick the show display and the fit (Fit, Fill or 1:1) in the right column under **Output**. Fit shows the whole picture, with bars if the screen's shape differs.
- **PREVIS** sets the NDI quality: previs is 960x540 at 30 fps (the default); Full is 1080p at 60. It only changes the NDI copy; the wall stays full rate. NDI can be off and the show carries on.

Mapping (Output › **Mapping...**)

The mapping page cuts the canvas into slices for LED walls or several displays. The default is one slice: the whole canvas onto the display.

- **Canvas** and **Display**: the picture size and which screen gets it. Changing the canvas size needs **Restart player**.

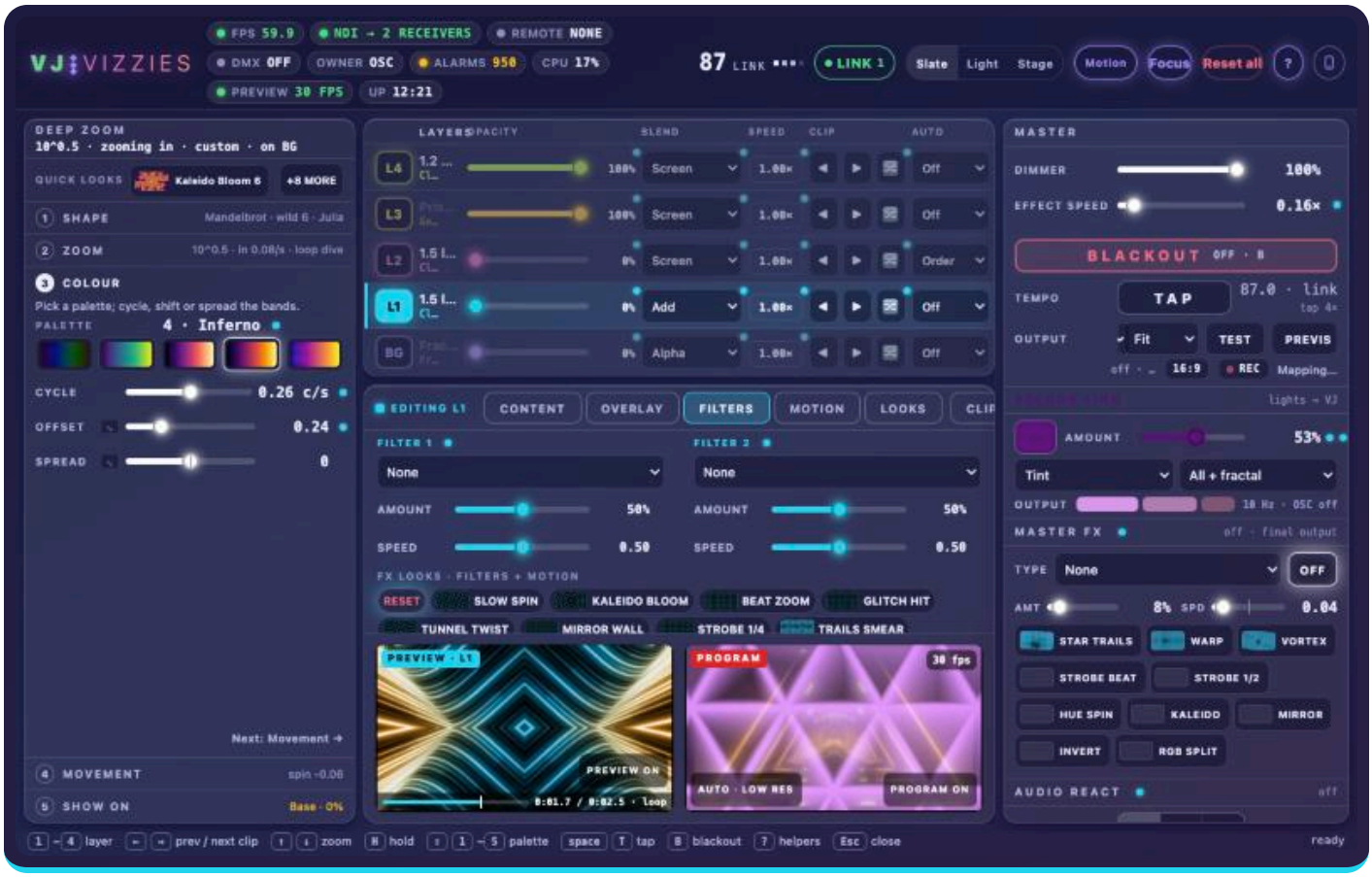
- **Cabinet** and **Test**: set your LED cabinet size, then show a display grid or a cabinet pattern to line up the wall.
- **Slices**: each slice takes a rectangle of the canvas (source X, Y, W, H) to a rectangle of the display, with rotation and flip. **Add** or **Duplicate** for more.
- **Clip scaling** per layer: how clips fill their layer (Cover crops to fill).
- **Venue**: save the whole setup as a preset and load it at the next show at that venue. Click **Save** to keep changes; **Revert** goes back.



The mapping page with the default single slice: the whole 1920x1080 canvas onto the HDMI display.

06 The control window

The control window has three columns: the fractal on the left, layers and the editor in the middle, master controls on the right. Open it from VJ VIZZIES.app or at `http://<this Mac>:8090`.



The whole control window. Monitors (PREVIEW and PROGRAM) sit under the editor.

Top bar, left to right

ITEM	WHAT IT TELLS YOU
FPS	Frames the player renders per second. 60 is normal.
NDI → N receivers	How many previs receivers are watching
REMOTE	Lights up while the iPad remote is connected
DMX	Whether the grandMA3 is sending
OWNER	Who moved things last: DMX, OSC or MIXED. The latest move wins.
ALARMS	Slow frames or sends since the player started. A few hundred right after a start is normal (the NDI helper warming up).
CPU	Player CPU, as a percent of one core

ITEM	WHAT IT TELLS YOU
PREVIEW	Frame rate of the monitors in this window
UP	How long the player has been running
BPM and LINK	Current tempo, its source, and Ableton Link peers. Click LINK to switch between Link and tap/DMX.
Slate / Light / Stage	Colour themes. Stage is dim red for dark venues.
Motion	Logo animation on or off
Focus	Dims the other layers while you edit one
Reset all	Clean start for every layer, after a confirm. Blackout, dimmer, output and scenes stay.
?	Helpers (hover tips) and Show OSC addresses



The Stage theme keeps the window dim on a dark stage.

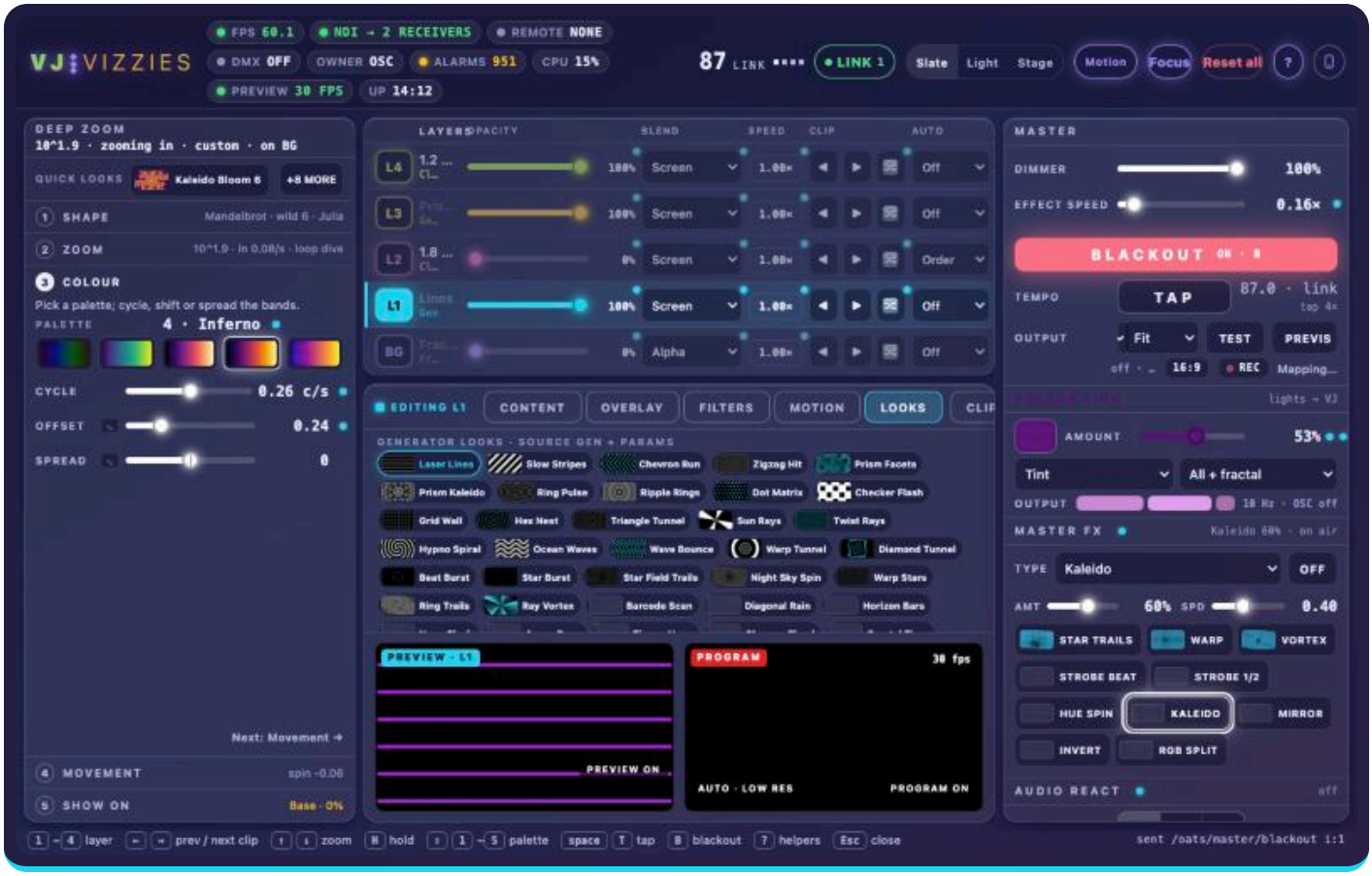
Middle column

- **Layers table**: one row each for L4, L3, L2, L1 and BG. Each row has Opacity, Blend, Speed, clip buttons (◄ ► shuffle), Auto (Off / Order / Shuffle) and Interval. Click a row to edit that layer.
- **Editor**: tabs Content, Overlay, Filters, Motion, Looks, plus Clips and Scenes. The label ("Editing L1") says which layer you're editing.

- **Monitors**: PREVIEW shows the selected layer alone; PROGRAM is the live output. Monitors appear when the window is at least 1280 px wide. Click PROGRAM to enlarge it.

Right column

- **Master**: Dimmer, Effect speed (0–2×), **BLACKOUT** (key B) and **TAP** tempo.
- **Output**: which display gets the show, Fit / Fill / 1:1, **TEST** pattern, **PREVIS** (NDI quality), record size and **REC**, and **Mapping…**.
- **Colour link**: lets the lighting colour tint the visuals.
- **Master FX**: one effect on the whole output. Pick a preset button; click **OFF** to clear it (clicking the preset again does not turn it off).
- **Audio react**: makes layers pump with a kick or bass from an input.
- **Test content**: Test card, NDI card, Off, Deep, Noise.



Blackout tested: B turns it on (PROGRAM goes black), B again brings the picture back.

Tips that work on every slider: click to jump, drag to fine-tune, double-click to reset, arrow keys to step. Hover any control to see what it does, its OSC address, and the grandMA3 channel that drives it.

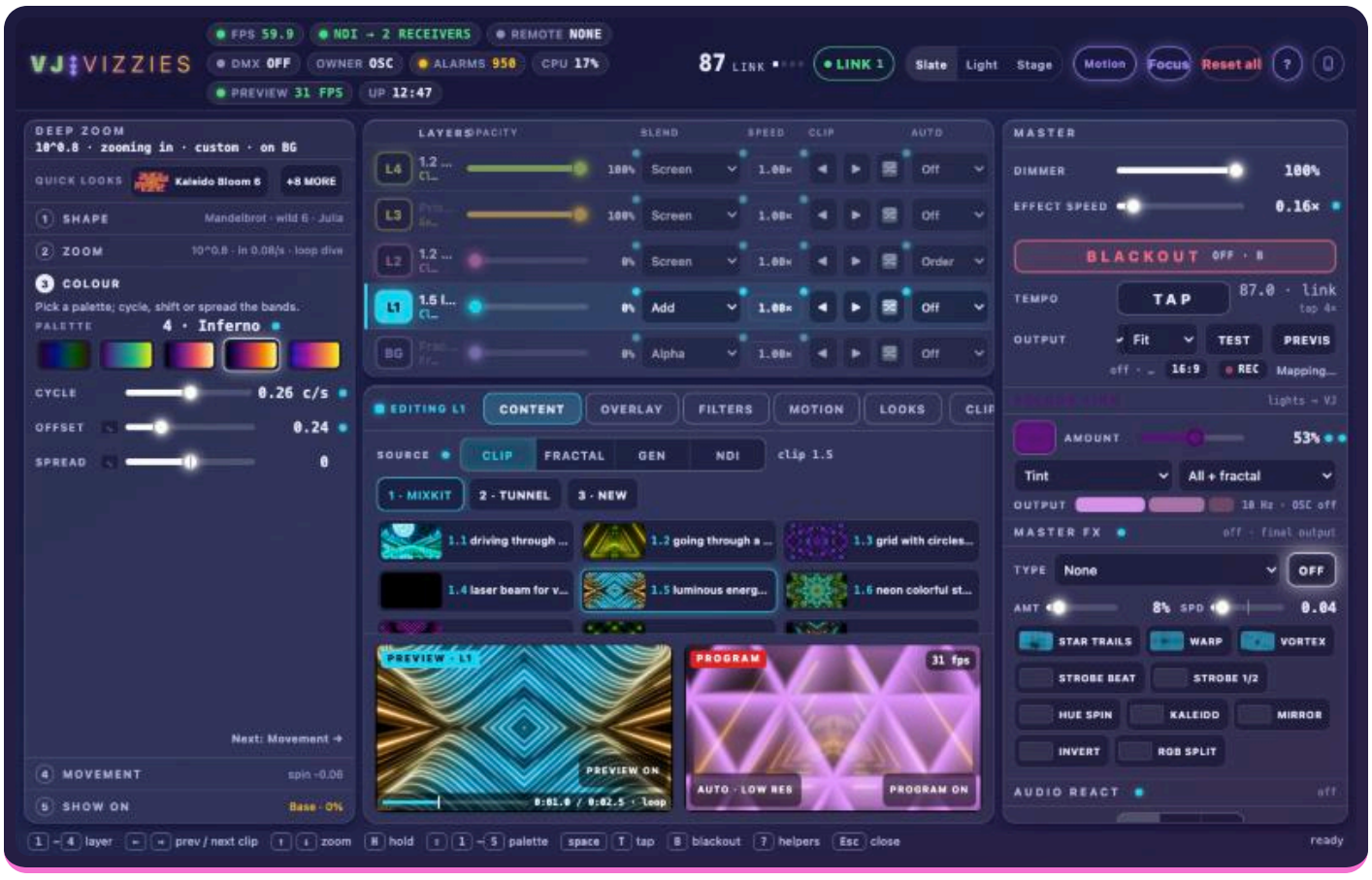
07 Layers, clips and the clip library

The picture is a stack: BG at the bottom, then L1, L2, L3 and L4 on top. Each layer shows one source, and its opacity and blend decide how it mixes with the layers under it.

The four kinds of source

(Content tab, top row)

SOURCE	WHAT IT SHOWS
CLIP	A video or still from the clip library
FRACTAL	The deep-zoom fractal. There is only one; it sits on whichever layer took it last.
GEN	A generated pattern: Lines, Angles, Prisms, Circles, Dots, Grid, Polygons, Rays, Spiral, Waves, Tunnel, Shapes Burst or Stars
NDI	A camera or another computer on the network, up to 4 at once. Green dot = live.



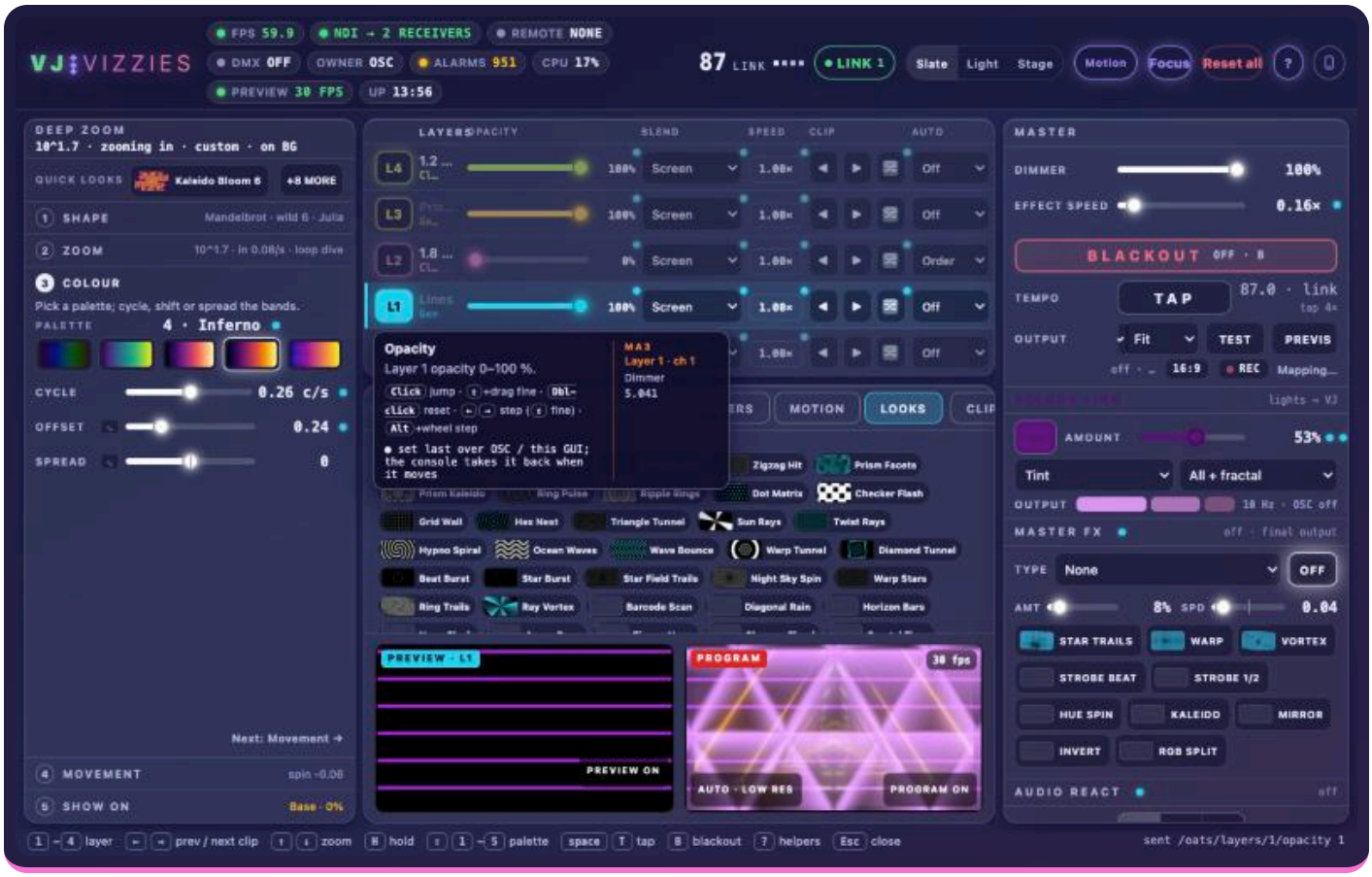
The Content tab for L1, showing the clip banks.

Bring a layer up in four steps

(tested on L1)

1. Click the layer's row in the Layers table (or press **1–4**, **0** for BG).
2. On the **Content** tab, pick CLIP, FRACTAL, GEN or NDI and choose what to show.

3. Drag its **Opacity** slider up. A layer at 0% is invisible, whatever is loaded.
4. Pick a **Blend**: Screen or Add brighten, Alpha covers, Multiply darkens.



> L1 at 100% with Laser Lines: the lasers show in PREVIEW and over the fractal in PROGRAM. Hovering the slider shows its tips and its grandMA3 channel.

The footer shows the command that went to the player, for example `sent /oats/layers/1/opacity 1`.

Playing clips

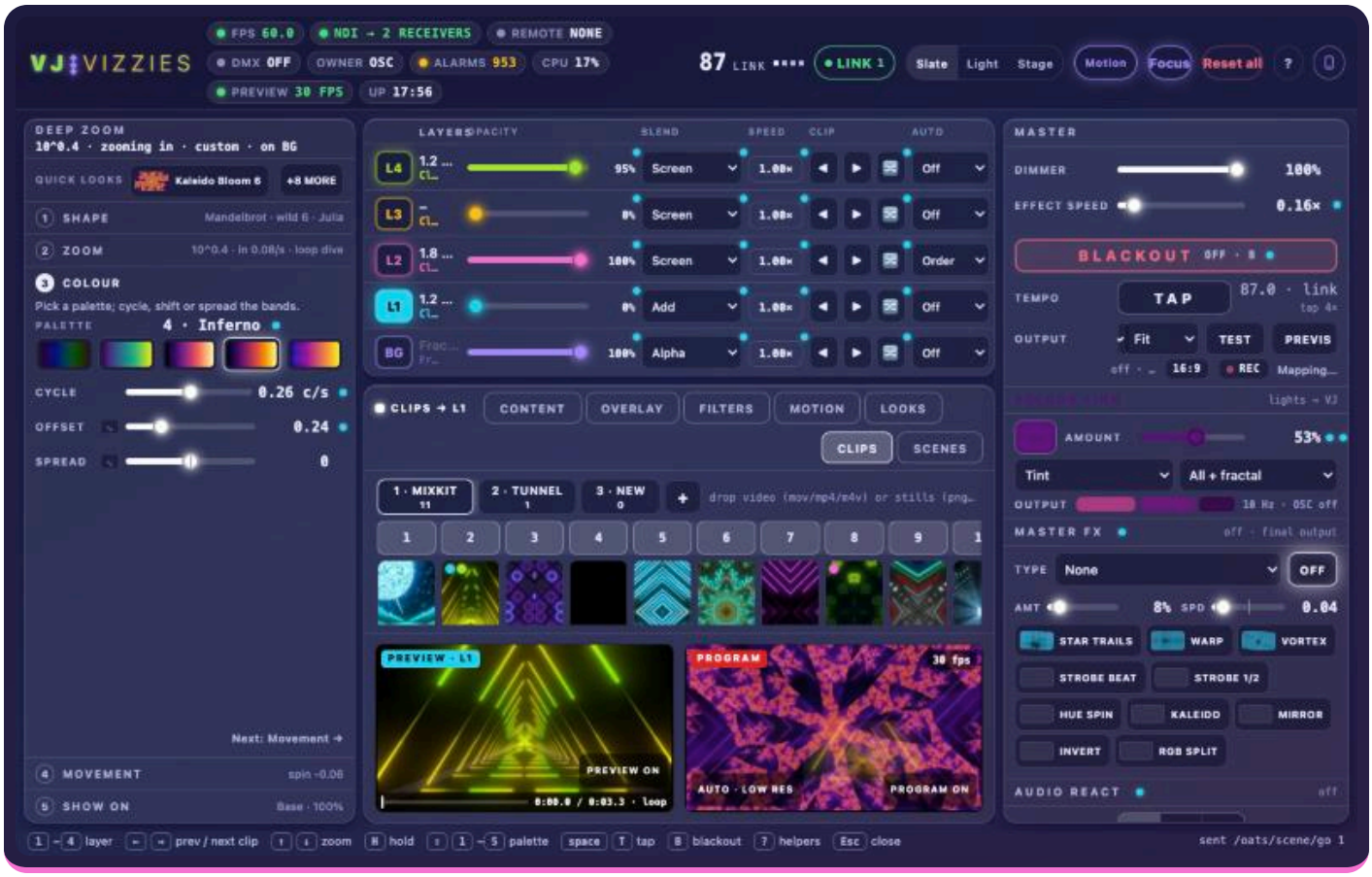
- ◀ ▶ steps to the previous or next clip; the shuffle button picks one at random. The ← → keys do the same on the selected layer.
- **Auto** plays through the bank by itself: Order or Shuffle, changing every **Interval** (¼ beat up to 32 beats, or 4 to 30 seconds).
- **Speed** runs from 0 to 4×, with a click-stop at 1×.
- On the Content tab, **Play mode** is Loop or Once, and **Transition** sets a crossfade of 0 to 10 s.
- ✕ clears the layer with a fade.

The clip library

(Clips tab)

- Clips live in banks. The suite ships three: **1 · MIXKIT** (11 clips), **2 · TUNNEL** and **3 · NEW**.
- Add clips by dropping a video (mov, mp4, m4v) or a still (png, jpg) onto a slot. **+** makes a new bank.
- Double-click a bank tab to rename it. Right-click a clip to remove it.
- You can also drag a clip, or a file from Finder, straight onto a layer row to play it there.

- On disk the library is `~/Applications/VJ VIZZIES/media/`. Updates keep it.



The Clips tab. Drop files onto a slot to add them.

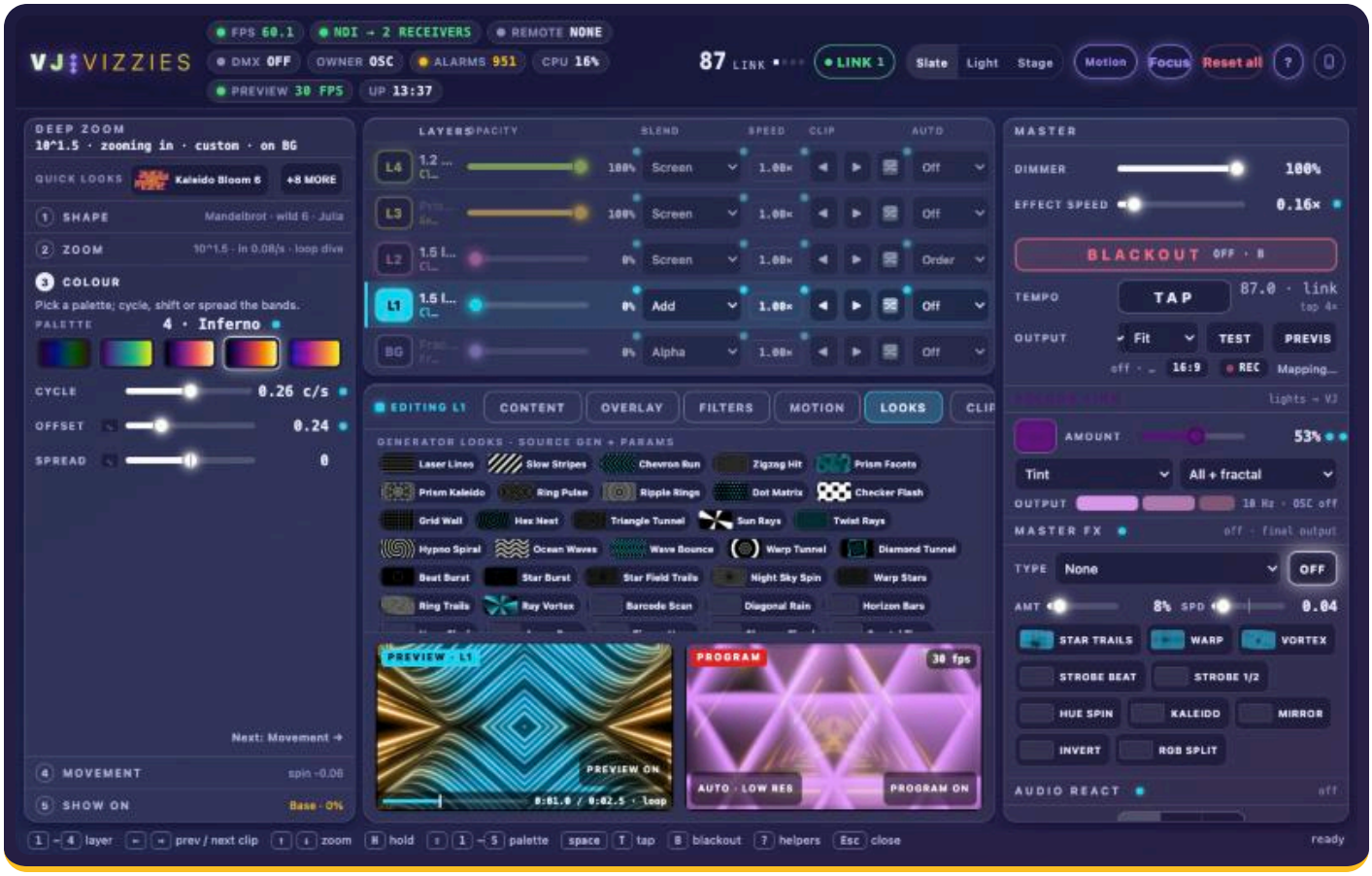
08 Generators, looks, shapes and overlays

A look is a ready-made generator setting: one click on the Looks tab turns the selected layer into it. There are 67 generator looks, and none of them need clips.

Looks tab

(tested: Laser Lines on L1)

- **Generator looks**: 67 buttons, for example Laser Lines, Slow Stripes, Chevron Run, Prism Kaleido, Hypno Spiral, Warp Tunnel, Star Burst, Ring Trails.
- **Scripts**: 5 timed sequences you start and stop with ▶ and ■: Geometry Tour, Build Up, Drop Hits, Chill Drift, Tunnel Ride.
- **FX looks**: preset filter combinations (see the FX chapter).



The Looks tab. One click turns the selected layer into that look.

Tune a generator by hand

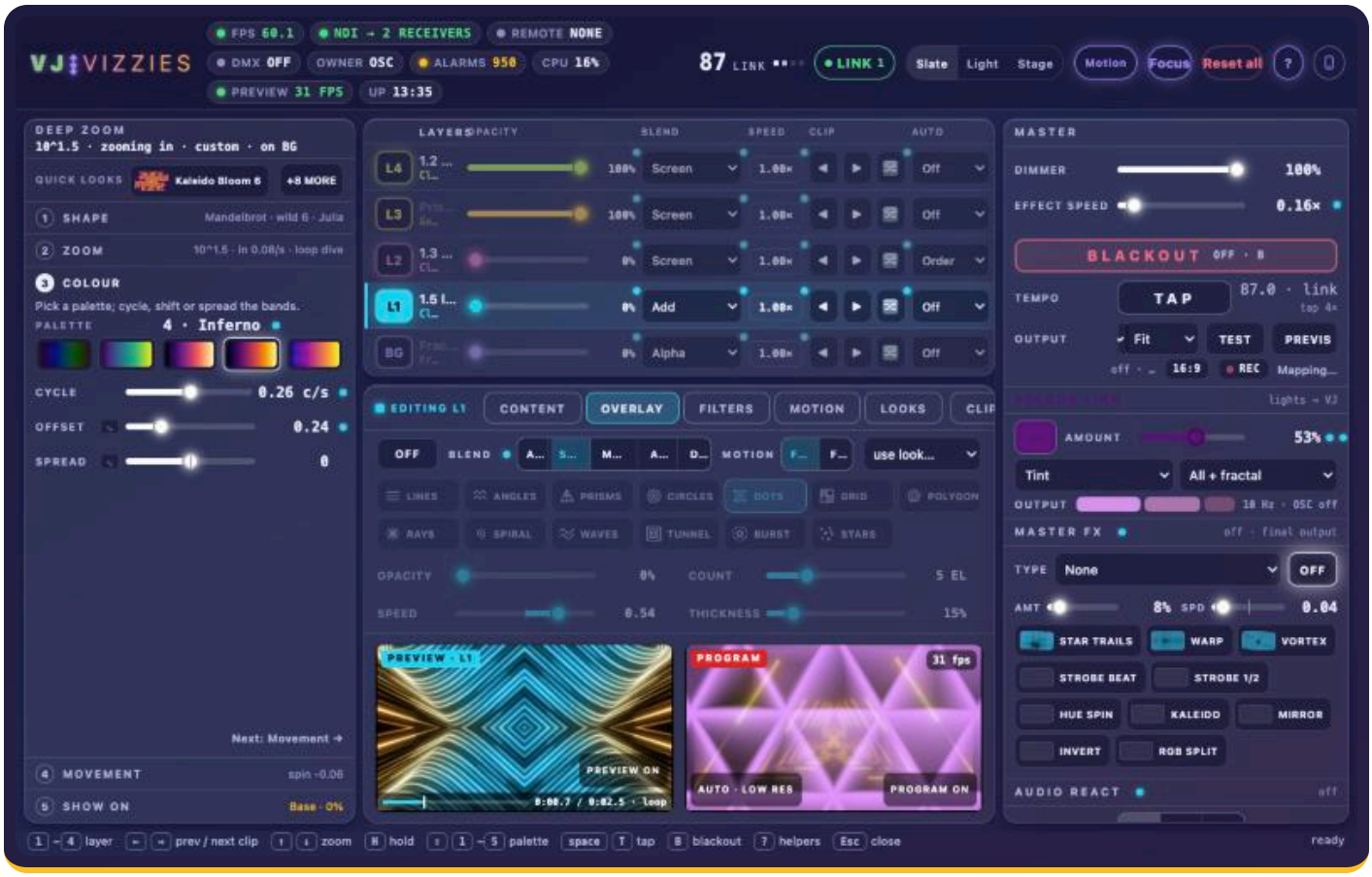
(Content tab → GEN)

- Pick one of the 13 shapes, then set Count, Thickness, Softness, Speed, Angle and Variation.
- **Colour**: White, Link (follows the lighting colour), Accent, or Custom with two colours.
- **Beat sync** locks its movement to the tempo.

Overlays

(Overlay tab)

- Every layer can carry a second pattern on top of its content: lines, angles, prisms, circles, dots, grid, polygon, rays, spiral, waves, tunnel, burst or stars.
- Set its own blend (Add, Screen, Multiply, Alpha, Diff), opacity, count, speed and thickness, or pick **use look...** to load a generator look as the overlay.
- **OFF** removes the overlay.



The Overlay tab.

The fractal

(Deep zoom card, left column)

The fractal is set up in five numbered steps, one open at a time. Each ends with a **Next** button.

1. **Shape**: Mandelbrot, Burning Ship, Tricorn, Celtic, Flowers, four Julias, Kaleido 6 and 8. Branches, Twist, Mirror and Detail shape it further.
2. **Zoom**: where to zoom, how deep (down to 10^28), In / Hold / Out, speed, and Loop (Off, Mini, Dive).
3. **Colour**: a palette plus Cycle, Offset and Spread. **Shift + 1–5** picks palettes from the keyboard.
4. **Movement**: rotation, spin and motion presets (Breathe, Pulse, Tide, Heartbeat, Mutate, Wobble).
5. **Show on**: which layer the fractal sits on, and its opacity.

Quick looks at the top of the card are 8 finished fractal scenes, such as Kaleido Bloom 6 and Burning Cathedral; **+8 MORE** opens the rest. The ↑ ↓ keys zoom and **H** holds.

09 FX, strobe and the beat clock

Every layer has two FX slots, and the whole output has one more (Master FX). All of them can lock to the beat clock.

The 20 effects

(the same list in every slot)

Kaleido, Mirror, Warp, Twist, Ripple, ZoomBlur, HueSpin, Posterize, Invert, Duotone, RGBSplit, Pixelate, SliceShift, Strobe, Blur, EdgeGlow, Trails, StarTrails, WarpTrails, SpiralTrails.

Layer FX

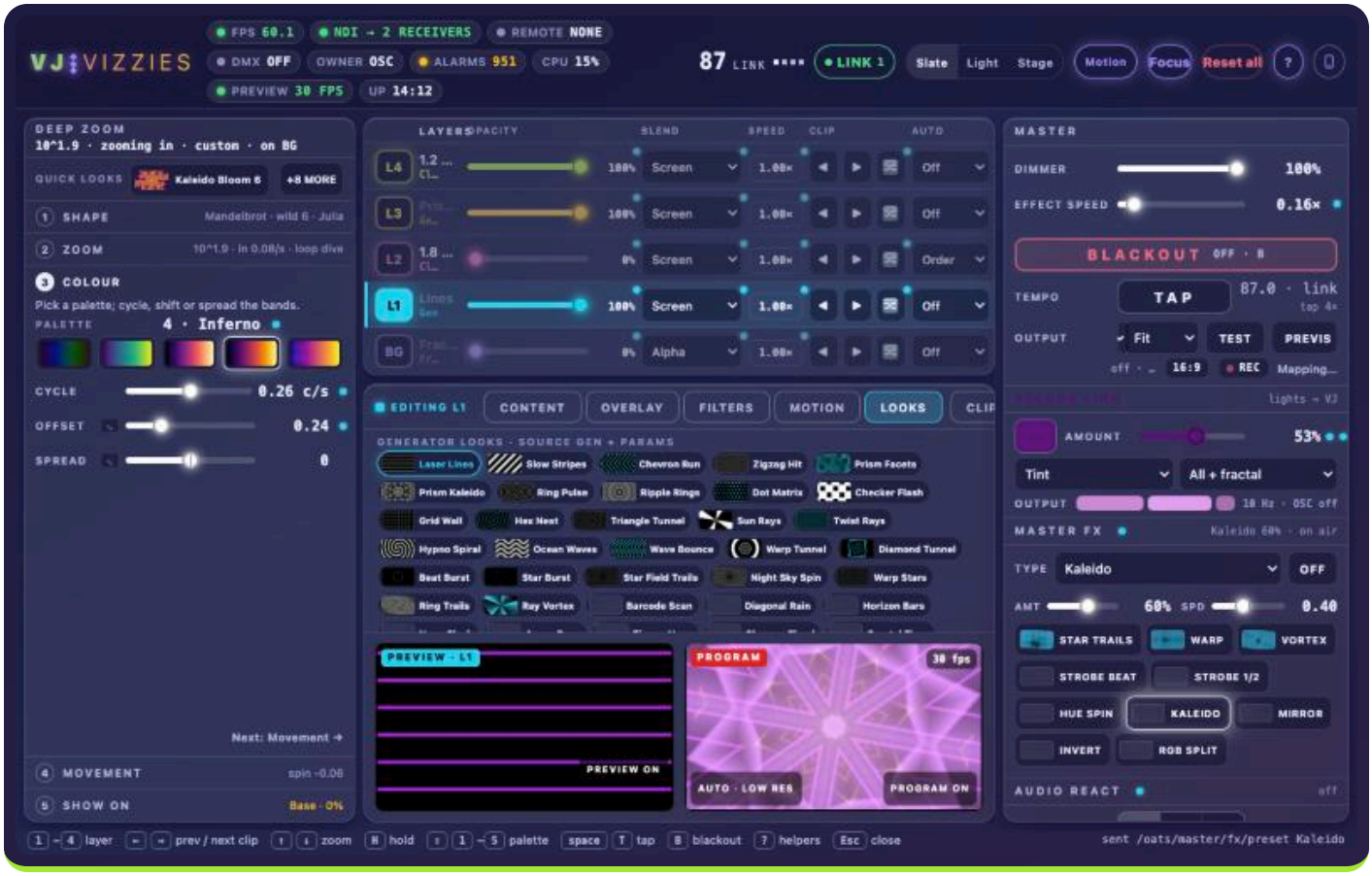
(Filters tab)

- **Filter 1** and **Filter 2**: pick a type, then set Amount and Speed.
- **FX looks** underneath are 47 ready-made combinations, such as Slow Spin, Kaleido Bloom, Beat Zoom, Glitch Hit, Strobe 1/4 and Trails Smear. **RESET** clears both slots.

Master FX

(right column; tested with Kaleido)

- Presets: Star Trails, Warp, Vortex, Strobe Beat, Strobe 1/2, Hue Spin, Kaleido, Mirror, Invert, RGB Split.
- AMT and SPD set strength and speed. **OFF** clears it.



Master FX tested: Kaleido turns the whole output into a kaleidoscope. OFF clears it.

The beat clock

- **Ableton Link** leads when it's on: start Link in djay (or any Link app) and the LINK chip turns green with its peer count. Click the chip to switch to tap or DMX.
- **TAP** (or the **Space** / **T** key): tap at least twice on the beat. It averages your last taps and resets after 2 seconds without a tap.
- The grandMA3 can also send BPM on the Master fixture, and OSC can set it with `/oats/tempo/bpm`.
- With no tempo at all, the player runs at 120 BPM.

Strobe speed follows the beat

SPEED SLIDER	FLASH EVERY
below 0.25	1 beat
0.25 to 0.5	1/2 beat
0.5 to 0.8	1/4 beat
0.8 and up	1/8 beat

It never goes over 4 flashes a second. At fast tempos it automatically drops to a slower division (see **Safety**).

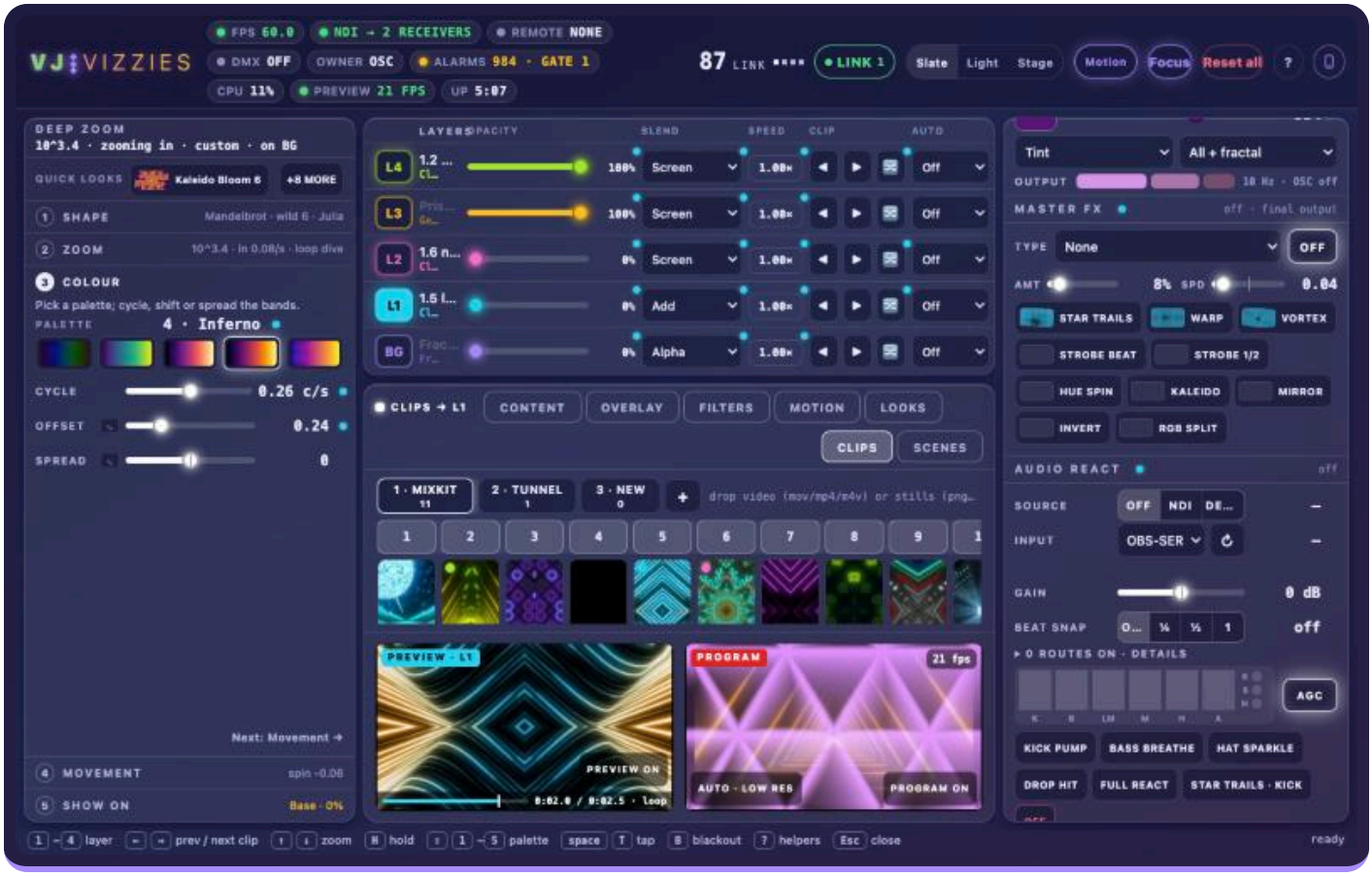
Beat mod (on each layer) moves one setting in time with the music: rotation, scale, position, FX 1, FX 2 or opacity. Its shape can be sine, saw, square or random, and its rate runs from 1/4 beat to 16 beats.

10 Audio react

Audio react listens to the music and moves the visuals with it: a kick can pump a layer, the bass can breathe the fractal, a drop can fire an effect. It lives in the right column under **AUDIO REACT**.

Set it up

1. **Source**: **NDI** to hear djay's Mac over the network, or **Device** for a sound card input. **Off** stops it.
2. **Input**: pick the djay Mac's NDI audio (or the device). The refresh button looks again for new sources.
3. **Gain**: raise it until the meter moves well. **AGC** keeps the level steady on its own.
4. Pick a preset: **Kick pump**, **Bass breathe**, **Hat sparkle**, **Drop hit**, **Full react** or **Star trails · kick**. **Off** clears them.



The AUDIO REACT card at the bottom of the right column, with the six-band meter and the presets.

Reading the meter: six bars for Kick, Bass, Lo-mid, Mid, High and Air, plus dots that flash on each kick, snare and hi-hat. For a big meter on another screen, open `http://<this Mac>:8090/vu`.

Fine control

- **Beat snap** lines hits up with the beat clock: off, ¼, ½ or 1 beat.
- **Details** opens the routes: up to 8, each one band or hit driving one setting (a layer's opacity, scale or spin, an FX amount, the fractal's zoom, the master dimmer, and more) by an amount you set.

- Kick detection listens at 20–120 Hz. When the DJ filters the kick out, hits stop instead of firing on leftovers, and come back smoothly when the kick returns.
- Over NDI, the card shows about 60 ms of delay from djay to the picture.

The card and its presets were checked in the control window; the music-driven behaviour is as documented in October and wasn't re-tested with live music for this guide.

11 Blend modes and busking stacks

A blend mode decides how a layer mixes with everything under it. Screen is the safe default for bright visuals on a dark background.

BLEND	WHAT IT DOES	GOOD FOR
Alpha	Covers what's underneath, by opacity	A clip that should replace the picture
Add	Adds light; bright parts get brighter	Lasers, lines, glows
Screen	Like Add, but softer and never clips to white	The everyday choice for stacking
Multiply	Darkens; black cuts holes	Masks, shadows, texture
Difference	Inverts where layers overlap	Glitchy, psychedelic looks
Lighten	Keeps the brighter of the two	Merging two bright layers
Darken	Keeps the darker of the two	Silhouettes
Overlay	Boosts contrast	Adding punch to a clip

Changing a blend

- Control window: the **Blend** drop-down on the layer's row.
- APC mini: **SHIFT + a layer's track button** steps to its next blend, or switch PAD MODE to **BLEND** to pick one directly.
- grandMA3: the Layer fixture's blend channel. OSC: `/oats/layers/1/blend s Screen`.

Stacks: whole looks in one hit

A stack sets several layers at once: content, blends, FX and opacity. There are 40, such as Cathedral Lasers, Laser Show and Drop: Strobe Grid. Fire them from the APC mini in PAD MODE **STACKS**. Layers a stack doesn't use go dark, so the result is always clean.

Busking recipe

1. Fire a stack for the base look.
2. Ride the layer faders to bring parts in and out.
3. Hold an **FX HIT** on the drop, and let go to snap back.
4. Hold a **BUMP** (spin, flip, invert, zoom) for a one-off move.
5. Store what you like as a scene.

12 Scenes: save and recall a look

A scene stores the whole look (every layer, the fractal and the FX) and brings it back with a fade. There are 8 banks of 16, so 128 scenes; "1.2" means bank 1, slot 2.

Store a scene

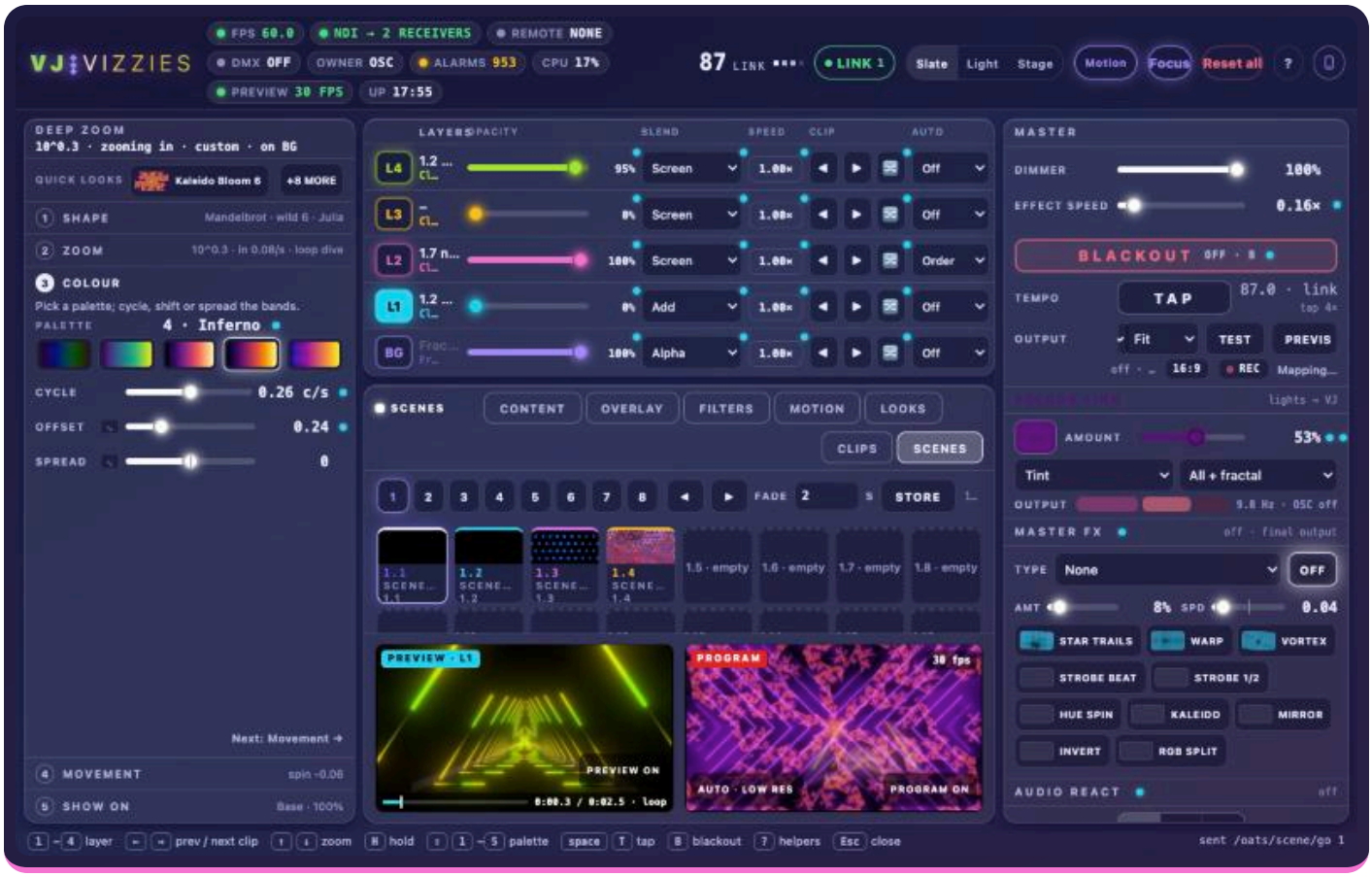
(Scenes tab)

1. Build the look.
2. Set **FADE** to how many seconds the recall should take.
3. Click **STORE**, then click an empty tile, and give it a name.

Recall a scene

(tested: scene 1.1 faded in over 2 s)

- Click its tile. It fades in over that scene's fade time.
- Right-click a tile (or long-press on a tablet) for more: **Go** (cut, no fade), Rename, Set fade, colour, Store again, Delete.
- ◄ ► and the numbered bank buttons move between banks.



Scene 1.1 recalled: PROGRAM faded to it over 2 seconds.

From other controls

FROM	HOW
APC mini	The 8 round scene buttons recall slots 1–8 of the current bank. SHIFT + a scene button picks the bank.
grandMA3	Master fixture channel 9 = scene number, channel 10 = fade time
OSC	`/oats/scene/go i 2` recalls scene 2; also `/oats/scene/next` and `/oats/scene/prev`

Not stored in a scene: blackout, dimmer and test content. A scene never blacks out the wall or unhides it.

Scenes are saved in `oats-m0/scenes.json`, with thumbnails beside the clips. Updates keep them.

13 Recording

REC records exactly what the wall shows (after master FX, dimmer and blackout) to a video file. HDMI and NDI carry on untouched while it records.

1. In the right column under **Output**, pick **16:9** (1920x1080) or **9:16** (1080x1920, a centre crop for vertical socials).
2. Click **REC**. It glows red and counts the time.
3. Click it again to stop.

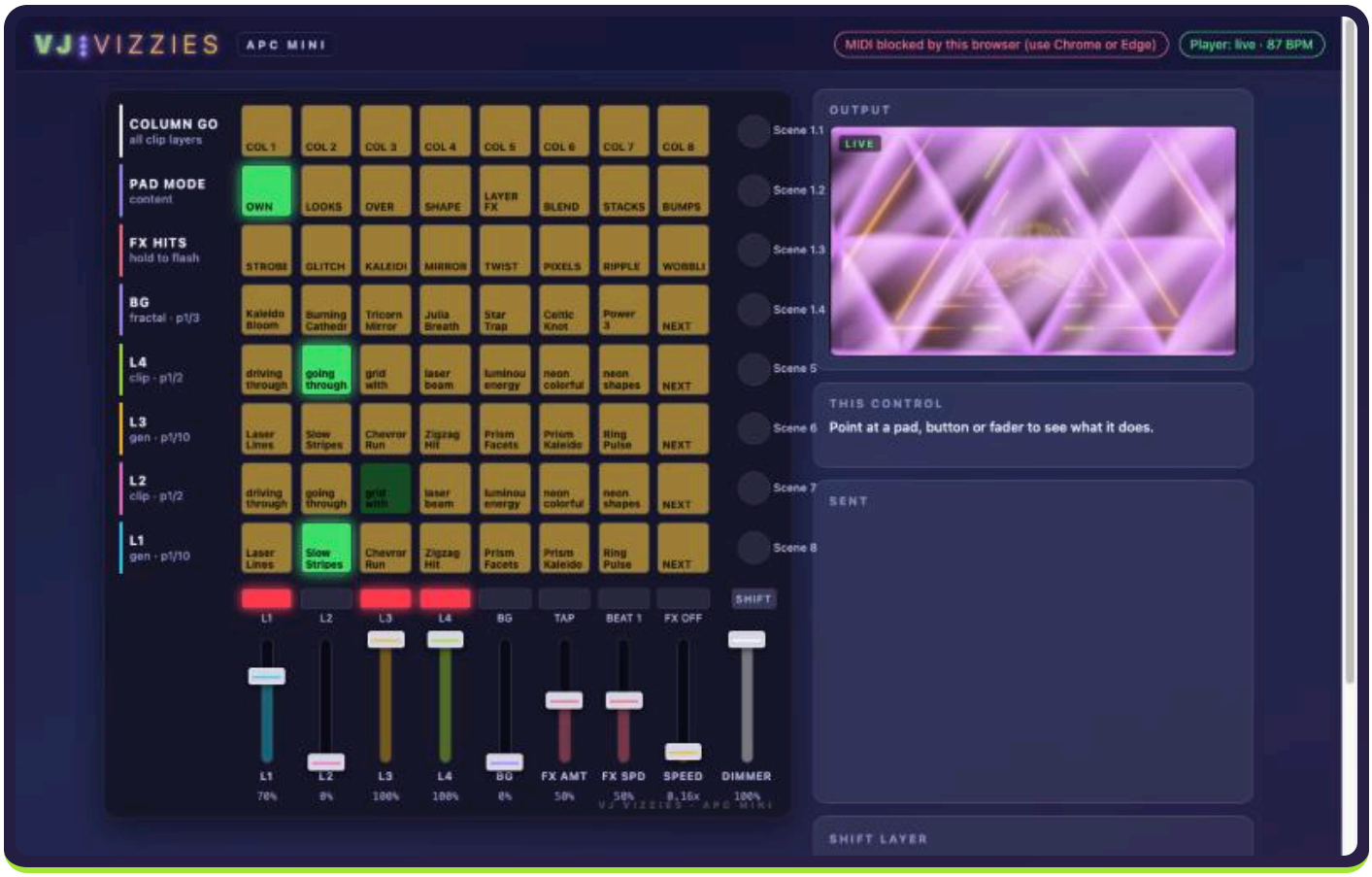
Where the files go: `~/Movies/OATS VIZ/teaser-takes/`, named by date and time and size, for example `20261106-021500-1080p.mp4`. They're H.264 at about 24 Mbit/s, at 60 fps, with no audio.

Disk guard: REC won't start with less than 3 GB free, and stops by itself at 2 GB free or after 10 minutes. Move finished files off the Mac.

While recording, keep the load light: avoid Auto clip changes on several layers at once, which can stall the player (see Troubleshooting). Over OSC: `/oats/record/start s 1080p` (or `vertical`) and `/oats/record/stop`.

14 The APC mini

The APC mini page is an exact on-screen copy of the Akai APC mini mk1. Click it with the mouse, or plug in the real APC mini and every pad, button and fader drives the player.



The APC mini page in OWN mode. The OUTPUT card on the right is a live view of the show.

Open it

1. Plug the APC mini into the Mac by USB.
2. Open **Chrome or Edge** at `http://127.0.0.1:8090/apc`. Safari and other addresses can't use MIDI.
3. The top bar says **APC mini connected** and **Player: live · BPM**. If it says *MIDI blocked by this browser*, you're in the wrong browser.

The grid, top row to bottom row

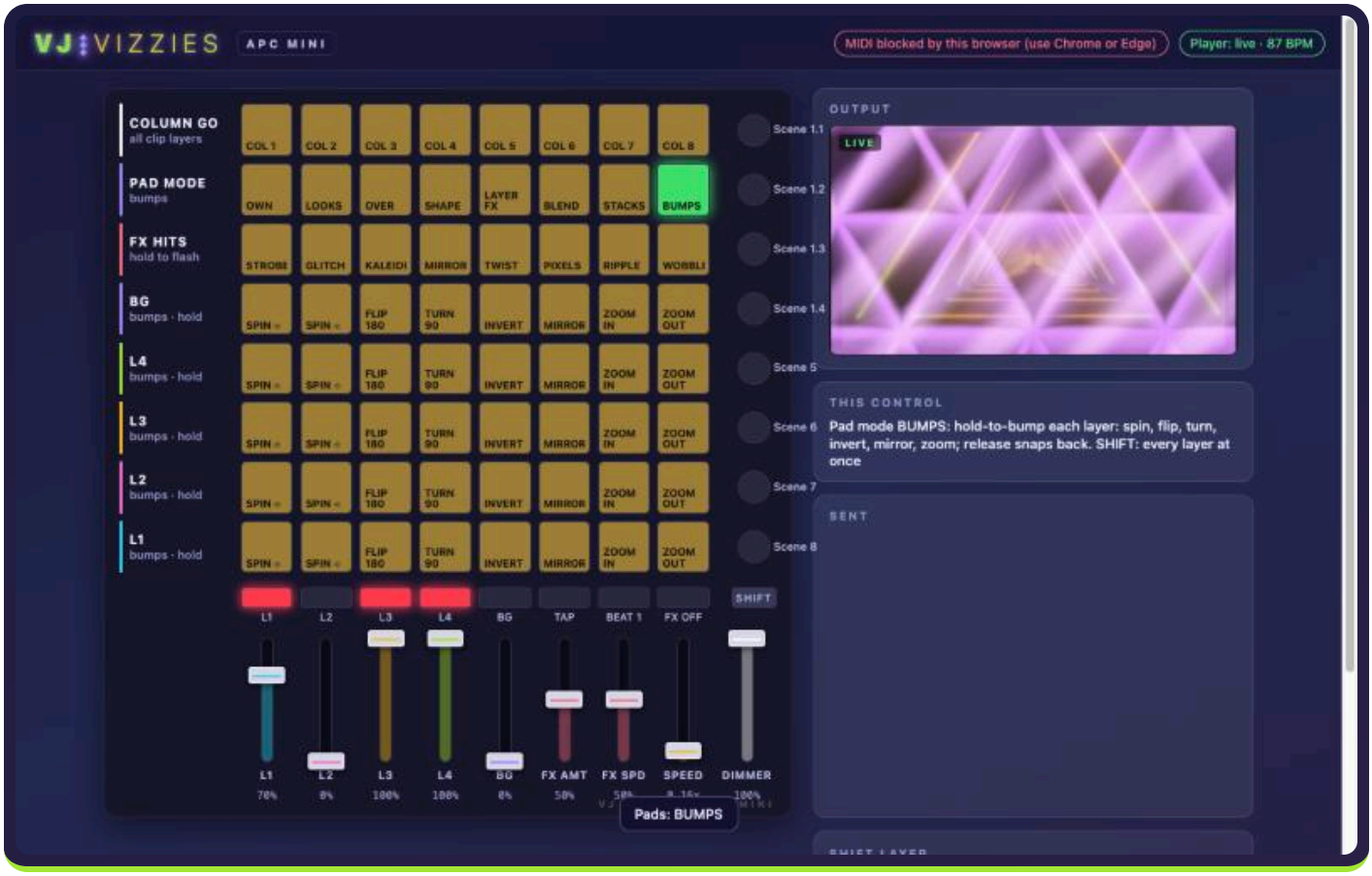
ROW	WHAT IT DOES
COLUMN GO	COL 1–8 plays that clip slot on every clip layer at once
PAD MODE	Picks what the five layer rows below do (next table)
FX HITS	Hold to flash a master effect, let go to snap back: STROBE, GLITCH, KALEIDO, MIRROR, TWIST, PIXELS, RIPPLE, WOBBLE
BG, L4, L3, L2, L1	One row per layer. Pads 1–7 are items; pad 8 is **NEXT**

PAD MODE: what the layer rows do

MODE	A PAD ON A LAYER ROW…
OWN	Plays that layer's own content: its clips, its fractal places or its looks
LOOKS	Turns the layer into a generator look
OVER	Puts a look on the layer as an overlay; tap the lit pad to remove it
SHAPE	Picks one of the 13 generator shapes (fractal shapes on the BG row)
LAYER FX	Applies an FX preset; tap the lit pad to clear it
BLEND	Picks one of the 8 blends
STACKS	Fires a whole multi-layer look (40 stacks)
BUMPS	Hold for a move that snaps back on release: SPIN ⟳, SPIN ⟲, FLIP 180, TURN 90, INVERT, MIRROR, ZOOM IN, ZOOM OUT



LOOKS mode: every layer row offers the same generator looks.



BUMPS mode: hold a pad for the move; let go and it snaps back.

Pad colours: green = playing now, yellow = ready, blinking = playing but the layer is at 0%. Firing a pad on a layer at 0% brings the layer up.

Buttons and faders

- **Track buttons** under the grid: L1–L4 and BG switch a layer on or off; **TAP** taps tempo; **BEAT 1** resets the downbeat; **FX OFF** drops all hits and clears Master FX.
- **Scene buttons** (round, right side): recall scenes 1–8 of the current bank.
- **Faders**: L1, L2, L3, L4, BG opacity, then FX AMT, FX SPD, SPEED (effect speed, middle = 1×) and DIMMER.
- Faders pick up softly: if the fader and the setting disagree, move the fader past the current value and it takes over without a jump.

SHIFT combos

(hold SHIFT; on screen, click SHIFT to latch it)

SHIFT +	DOES
pad 8 on a layer row	Next page of 7 items
an FX HIT	The other 4 hits: DUOTONE, STAR TRAILS, WARP, VORTEX
a scene button	Picks scene bank 1–8
a layer's track button	Next blend for that layer
a BUMP	Bumps every layer at once
BEAT 1	Standby logo card on or off
TAP	Follow or ignore Ableton Link
FX OFF	Blackout

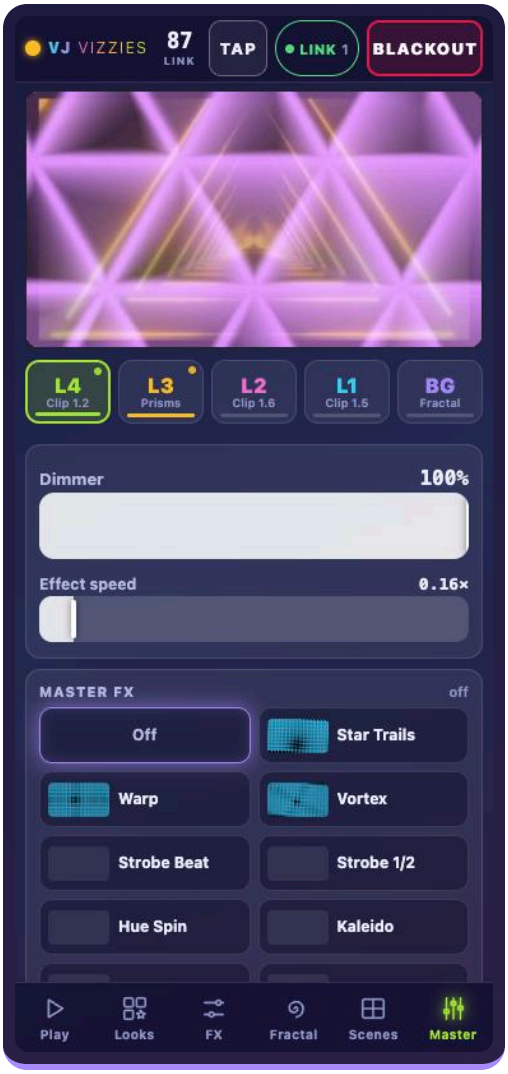
The side cards: OUTPUT is a live view of the show; THIS CONTROL explains whatever you point at; SENT logs every command; SHIFT LAYER is a reminder of the combos above. Tested: Slow Stripes on L1 and an L1 INVERT bump both reached the player.

15 Phone page and iPad remote

A phone and an iPad can both run the show from the stage. They go through the same control window server, with the same network rules and PIN, so they can't reach the player any other way.

Phone: open `http://192.168.1.148:8090/phone` (a phone opening the main address is sent there).

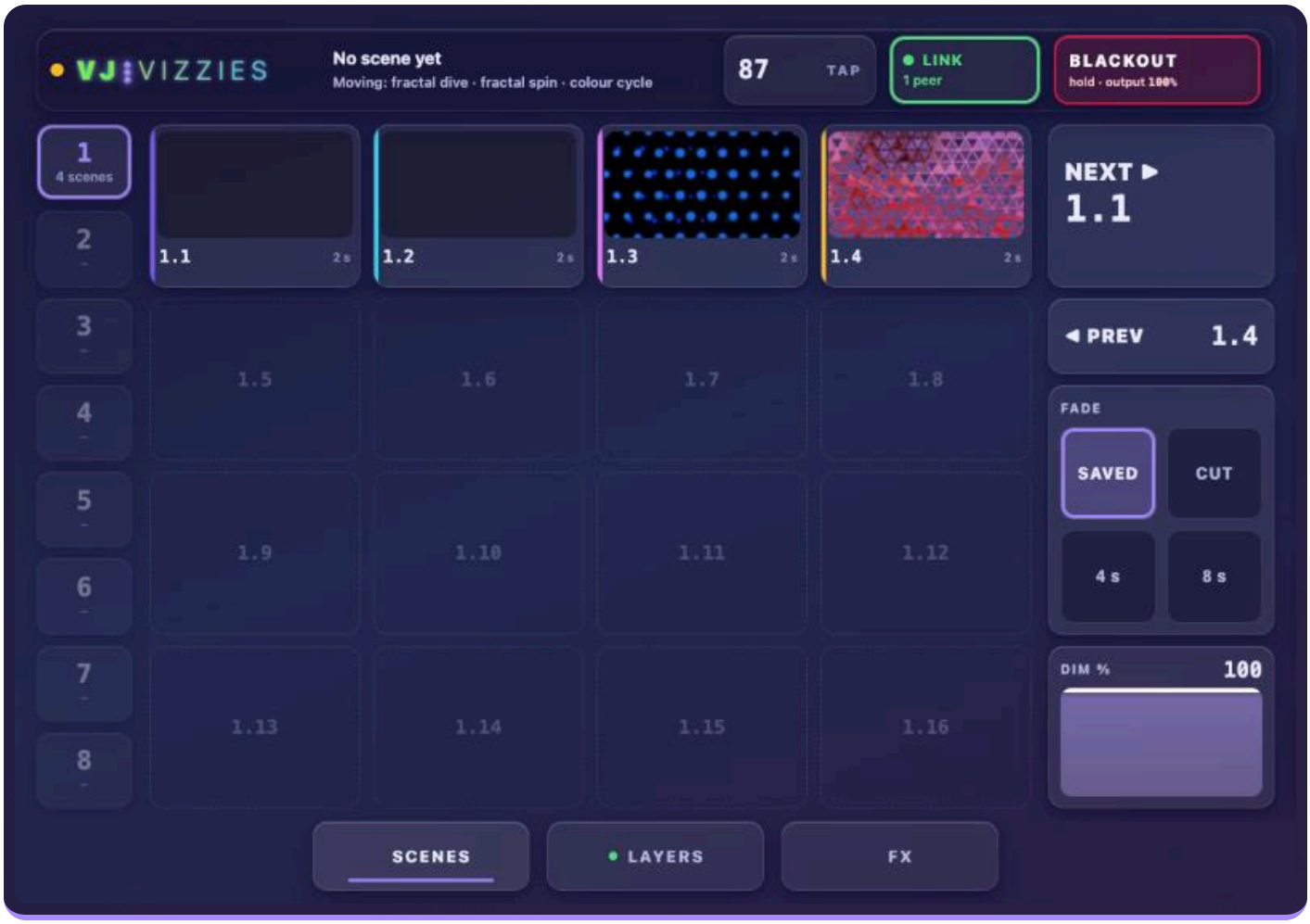
- Along the top: the health dot, BPM, **TAP**, **LINK** and **BLACKOUT**.
- A live preview of the output, then one chip per layer (L4 to BG) showing what's on it.
- Tabs at the bottom: **Play**, **Looks**, **FX**, **Fractal**, **Scenes** and **Master**. The Master tab has the dimmer, effect speed and the master FX buttons.



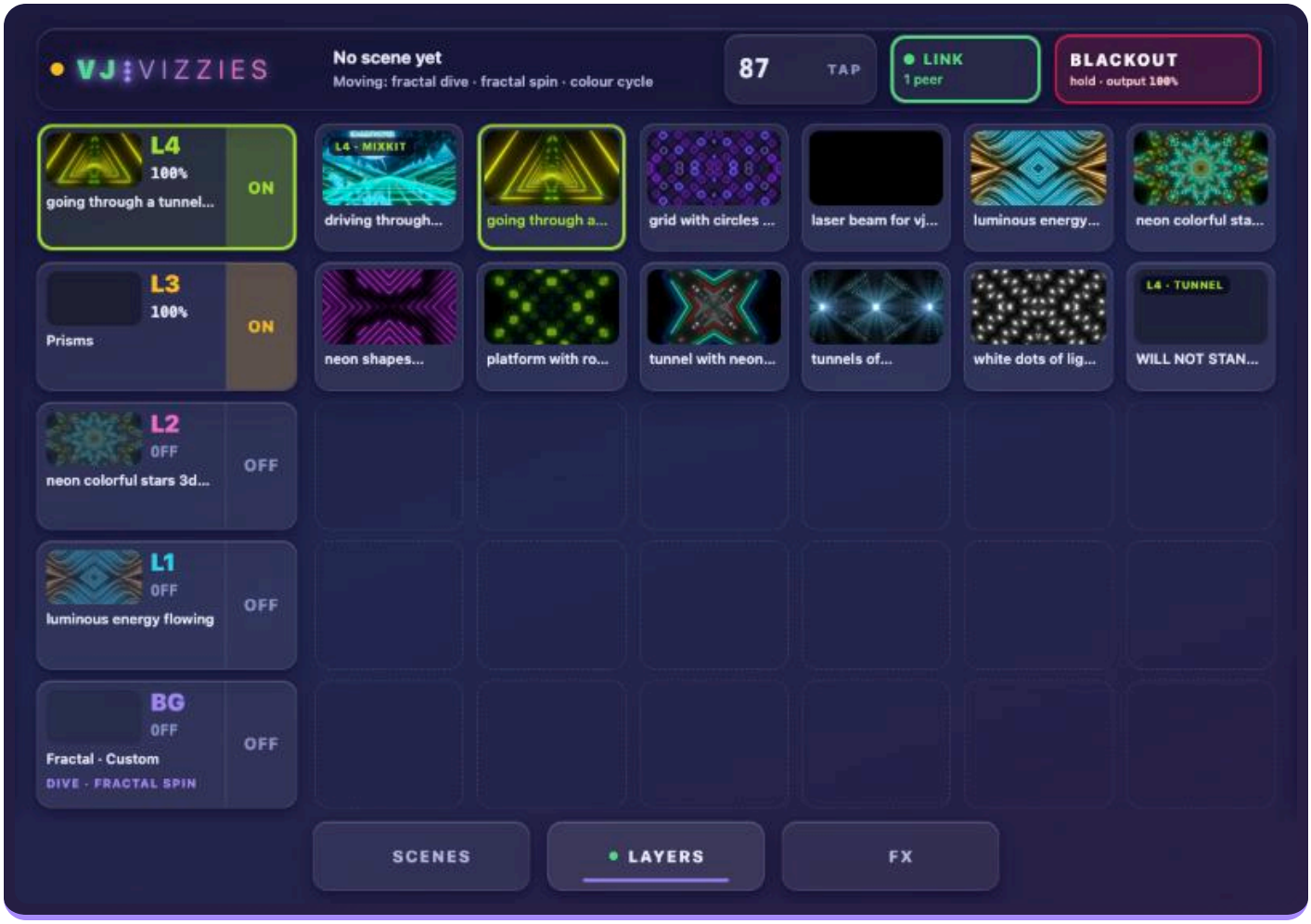
The phone page on the Master tab, with the live preview at the top.

iPad remote: open `http://192.168.1.148:8090/remote` in Safari, then Share › **Add to Home Screen** so it opens full screen. Three pages, chosen with the buttons at the bottom:

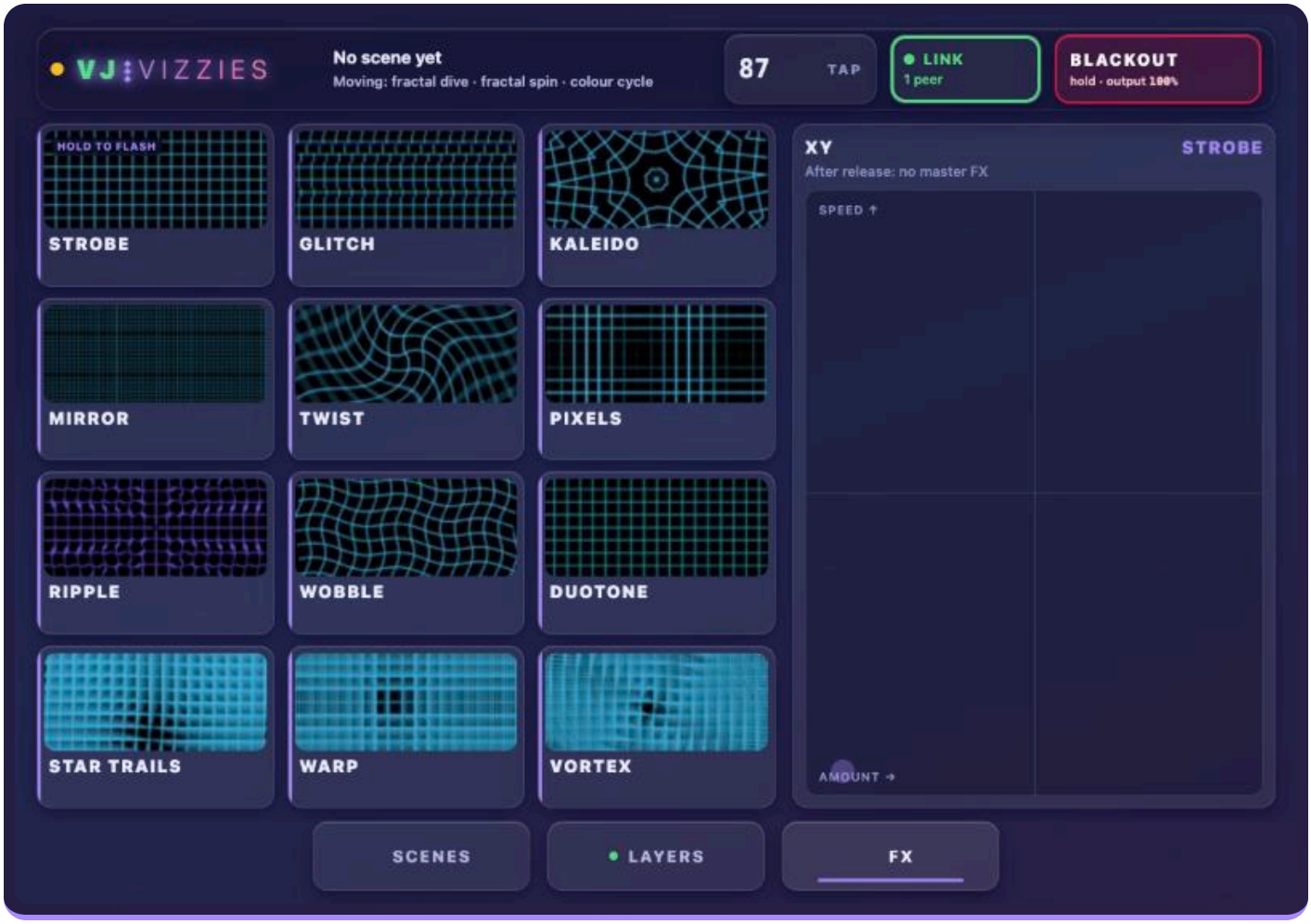
PAGE	WHAT'S ON IT
Scenes	Banks 1–8 down the side, a 4x4 grid of scene pads, **NEXT** and **PREV**, the fade choice (**SAVED**, **CUT**, 4 s, 8 s) and the dimmer
Layers	A tile per layer with ON/OFF, and a grid of that layer's clips or looks. Firing a pad on a layer that's off turns it on
FX	12 hold-to-flash hits (Strobe to Vortex) and an XY pad: amount across, speed up. On release the master effect goes back to what it was



Scenes: bank 1 holds four scenes; the right side picks the fade and the dimmer.



Layers: L4 is selected, so the grid shows its clips; L3 runs the Prisms look.



FX: hold a hit to flash it; the XY pad sets its amount and speed.

The bar on top of every page: **TAP** with the BPM, **LINK** (hold for a second to switch), and **BLACKOUT** (hold for half a second to fire, tap to release). The control window's **REMOTE** light shows when an iPad is on the page.

Both must be on the Netgear's Wi-Fi, not the building's. Checked for this guide: all pages load and show the live state (screens below); no pads were fired.

16 Keyboard shortcuts

These work in the control window whenever you're not typing in a field.

KEY	DOES
1 – 4	Select layer L1–L4 and open it in the editor
0	Select BG
← / →	Previous / next clip on the selected layer
↑ / ↓	Fractal zoom in / out
H	Hold the zoom
Shift + 1 – 5	Fractal palette: OATS Navy, Viridis, Magma, Inferno, Plasma
Space or T	Tap tempo
B	Blackout on / off
?	Helpers on / off
Esc	Close menus, pop-ups and the enlarged preview

On any slider: click jumps to the value; Shift- or Alt-drag is 10× finer; double-click resets; ← → step (Shift for fine); Alt + scroll wheel steps.

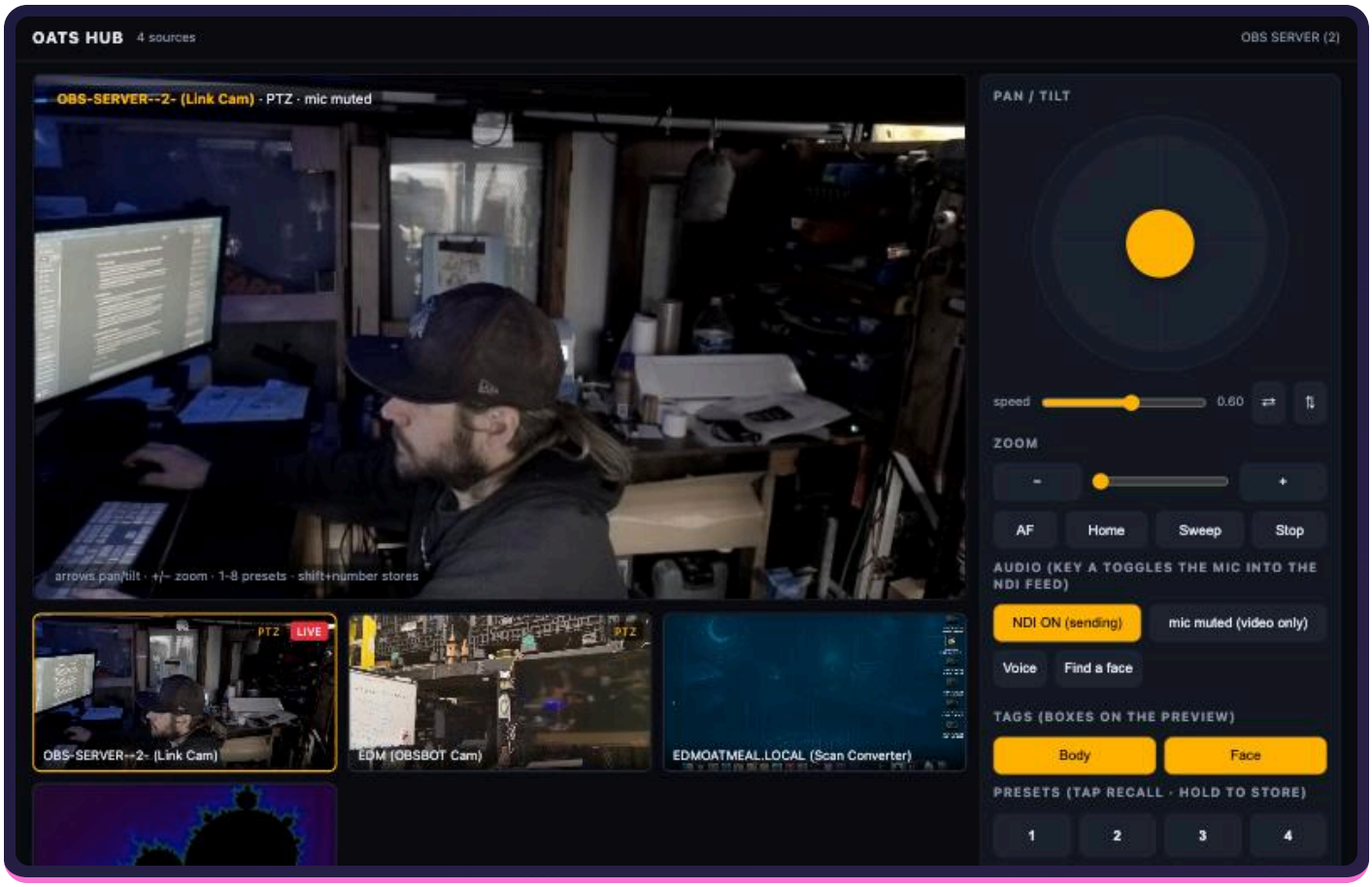
Camera hub page: arrows pan and tilt, + / – zoom, 1–8 recall presets (Shift + number stores), A toggles the mic, V voice, Space stops.

17 Cameras: the hub and OATS CAMS

The hub turns USB cameras into NDI sources and gives you pan, tilt, zoom, tally and face tracking in a browser. Open it at `http://<this Mac>:7080`.

The hub page

- **Big view** of the selected camera, with a strip of every source under it (PTZ, MIC and LIVE badges). Click a tile to select it.
- **PAN / TILT** joystick, a speed slider, and ⇄ / ⇅ to invert the axes.
- **ZOOM** – / +, plus **AF** (autofocus), **Home**, **Sweep** (slow pan) and **Stop**.
- **AUDIO**: **NDI ON** sends the camera; the mic button puts the mic in or out of the NDI feed; **Voice**; **Find a face** starts tracking.
- **TAGS**: Body and Face boxes drawn on the preview.
- **PRESETS 1–8**: tap to recall a camera position, hold to store one.



The hub with the Link Cam selected.

Other hub pages

- `/multiview`: a wall of every source. F = full screen, 1–9 = columns.
- `/tally`: a phone becomes a red / green tally light for one camera.

OATS CAMS (menu bar, house icon)

- Shows each camera: running, not connected, frozen, or tracking.
- **Restart cameras** fixes a camera that came up frozen. **Restart hub** restarts the page and multiview.
- **Open Hub**, **Open Multiview**, **Open logs folder**, **Open at login**, **Check for Updates...**.
- Quitting OATS CAMS leaves the cameras running.

Camera rules

- One camera per USB port.
- Close OBSBOT Center and Insta360 Link Controller: they fight the hub for control.
- Any player layer can show a camera: Content tab → **NDI**, then pick it.

18 Driving it from grandMA3 and OSC

The grandMA3 runs the player like a fixture on sACN universe 5, so visuals live in the same cues as the lights. OSC on port 7000 does the same from any other app.

Set up the grandMA3

1. Import `gdtf/OATS@OATS VIZ.gdtf` (in the VJ VIZZIES folder) into the fixture library.
2. Patch it on universe 5 at the addresses below, and send sACN to the Mac's lighting network.
3. The control window's **DMX** pill lights up when data arrives.

FIXTURE	ADDRESS	CONTROLS
Master	5.001	Dimmer, BPM, tap, beat resync, column go, scene + fade, effect speed, test content, blackout
Fractal	5.017	Shape, zoom target and depth, palette, colour cycle, rotation, spin
Layer 1–4	5.041 / .057 / .073 / .089	Opacity, clip bank + slot, blend, speed, transition, source, generator
Color	5.105	The lighting colour that tints the visuals
Layer FX 1–4	5.113 / .137 / .161 / .185	Position, scale, rotation, FX 1 + 2, beat mod (Layer FX 1 also holds master FX)

Who's in charge? The latest move wins, per setting: move a fader on the console and it takes that setting back from the control window. If DMX stops for 2.5 s, the player holds the last look.

OSC

(UDP 7000, from the lighting network or the Mac itself)

Tip: in the control window, **? → Show OSC addresses** labels every control with its address, and the footer shows each command as you click.

TO DO THIS	SEND
Layer opacity	`/oats/layers/1/opacity f 0.8` (layer 0 = BG)
Play a clip	`/oats/layers/1/play i 1 i 3` (bank 1, slot 3)
Next / previous clip	`/oats/layers/1/next`, `/oats/layers/1/prev`
Blend	`/oats/layers/1/blend s Screen`
Layer FX preset	`/oats/layers/1/fx/preset s "Slow Spin"`
Master FX	`/oats/master/fx/preset s Kaleido`, clear with `/oats/master/fx/reset`
Recall a scene	`/oats/scene/go i 2`
Tempo	`/oats/tempo/bpm f 128`, `/oats/tempo/tap`, `/oats/tempo/resync`
Blackout	`/oats/master/blackout i 1` (0 = off)
Standby / test card	`/oats/test/content s standby` (also `test`, `ndi`, `normal`)

19 Safety: strobe cap, standby and test card

The Strobe FX never flashes faster than 4 times a second, whatever the tempo, to protect people with photosensitive epilepsy.

The strobe cap

- Strobe speed follows the beat clock: the slider picks 1 beat, 1/2, 1/4 or 1/8 per flash.
- If the tempo would push it past 4 flashes a second, it halves the rate automatically. With no tempo set, it runs on a 120 BPM clock.
- The cap covers the **Strobe** FX only. A fast opacity flicker you build yourself (for example with the beat-mod LFO) is not capped, so keep those slow.
- The UK HSE guidance is 4 flashes a second; web guidance (WCAG) is 3. When in doubt, go slower.

The standby card

- The VJ VIZZIES logo on a moving navy background. It's what the wall shows at a cold start.
- Use it between sets, during changeovers, or whenever you need the wall to look intentional but quiet.
- The control window has no Standby button. Toggle it from the APC mini with **SHIFT + BEAT 1**, or over OSC with `/oats/test/content s standby` (back with `normal`).

The test card

- Colour bars, greyscale, a grid, a frame counter and a time-of-day clock, for lining up the LED processor or projector.
- Show it from the control window: right column, Test content › Test card (Off brings the show back). The Output row's TEST button shows a line-up pattern on the show display.

20 Security

Only the show Mac itself and the lighting network can use the control window, the previews, OSC and sACN. Everything else is refused, and refused addresses are logged.

- **PIN**: the control server can require a PIN (set `OATS_GUI_PIN` where the server starts). Then every page, phone and iPad included, asks for it, and five wrong tries lock the login for 60 seconds. It's **off** on the show Mac today. Turn it on for shows where other people can reach the lighting network.
- **Restarts** from the control pages are limited to one a minute.
- **Never**: join the show Mac to the Netgear's Wi-Fi; install a build or update mid-show; let a bot or AI change code on the show Mac (bots may only build and run looks).

21 Troubleshooting

Most problems are a wrong browser, a stopped player, or a camera that needs a restart.

The status lights are the dashboard. Read them across the top of the control window:

LIGHT	GOOD	LOOK INTO IT WHEN
FPS	59.5 to 60	Below 59.5 something heavy is running; below 50 a stall may be coming
NDI	Names a receiver	"no NDI receiver": Vision hasn't picked the source
REMOTE	iPad	none while the iPad should be on
DMX	Live	Off while the grandMA3 should be sending (see the grandMA3 chapter)
ALARMS	Steady	Climbing fast: the NDI helper is struggling. A few hundred right after a start is normal
CPU	Under about 20%	High: a build, transcode or busy browser is running on the Mac
LINK	A peer	No peer: djay isn't running or isn't on the network

Common problems

PROBLEM	FIX
macOS won't open the app the first time	System Settings > Privacy & Security > **Open Anyway**. Once per Mac.
Red "Player offline" banner in the control window	The player stopped. Menu-bar dot > **Start**, or `./m0.sh status` to see why.
Wall shows the standby logo	That's the cold-start card. Bring up any look, or SHIFT + BEAT 1 on the APC.
APC page says MIDI blocked	Use Chrome or Edge at `http://127.0.0.1:8090/apc`, not Safari and not the Mac's network address.
APC fader does nothing	Soft pickup: move it past the current value first.
A layer shows nothing	Its opacity is 0%, or the layer is off. Check the Layers table.
Master FX won't turn off	Click **OFF** next to the type, not the preset again.
NDI layer goes black with an amber note	The source stopped sending. Try **Proxy** bandwidth on a busy network.
A camera is frozen	OATS CAMS > **Restart cameras**.
Camera fights back on PTZ	Close OBSBOT Center and Insta360 Link Controller.
Two Macs, only one shows in Vision	Each Mac needs its own NDI machine name (Set up / Update this Mac...).
No DMX	Another app on the Mac may be taking sACN (sACNView, MA3 onPC). Close it.
Look lost after a restart	Recall a scene, or use `oats_state.py restore`.
NDI goes over Wi-Fi	Don't join the show Wi-Fi from the Mac when it's on the lighting cable.

PROBLEM	FIX
Preview looks jerky	Close extra control windows; the monitors run at 30 fps and drop frames rather than lag.

When the player restarts on its own. If the picture freezes for more than a second, a watchdog restarts the player and the wall goes black for about 4 seconds. It comes back with an empty layout, not your look, so recall a scene straight away. Keep a "safe" scene in bank 1 for this.

Fast Auto clip changes can cause it. While writing this guide, a layer set to Auto every 2 beats (a new clip about every 1.4 seconds) stalled the player twice, even with that layer at 0%. Until that's fixed, keep Auto at 4 beats or slower, and turn Auto off on layers you aren't showing.

22 Show-day checklist

Run this before doors; it takes about five minutes once you know it.

Before doors

- [ ] Show Mac on the lighting network by cable (192.168.1.148); its Wi-Fi not on the Netgear
- [ ] VJ VIZZIES running: menu-bar dot green, standby card on the wall
- [ ] Output on the right display with Fit; TEST pattern on the wall, then off
- [ ] Vision shows **OBS-SERVER--2- (OATS VIZ Program)**; the NDI light names the PC
- [ ] grandMA3 sending sACN on universe 5; DMX light live
- [ ] djay running with Link on; LINK shows a peer and the BPM matches
- [ ] Audio react source set and the meter moving (if you use it)
- [ ] APC mini page open in Chrome or Edge; "APC mini connected"
- [ ] iPad remote on the Netgear's Wi-Fi; REMOTE light on
- [ ] A safe scene in bank 1, fired once and checked on the wall
- [ ] No Auto faster than 4 beats; Auto off on hidden layers
- [ ] At least 10 GB free if you'll record
- [ ] PIN on if strangers can reach the lighting network

During the show

- [ ] Glance at FPS and ALARMS between songs
- [ ] One trails effect at a time
- [ ] No updates, builds or second player; no pile of browser tabs on the show Mac

After

- [ ] Store the looks you liked as scenes
- [ ] Move recordings off the Mac
- [ ] Write down anything that went wrong

23 Explainer video plan

Shoot the video as 10 short chapters of 1 to 3 minutes, about 20 minutes in all, each following one chapter of this guide. Short chapters are easier to reshoot and to post as clips.

#	CHAPTER	ON SCREEN	SAY / SHOW
1	What it is	The wall running a look, then the control window	Layers + fractal + FX, out HDMI; controls from a browser, APC or grandMA3
2	Install	The dmg, the drag, Open Anyway, the two questions	Two questions, clips included, updates keep your stuff
3	The rig	The rig diagram in this guide	HDMI is the show, NDI is previs only
4	Control window tour	Full window; point at each column	Top bar health, layers in the middle, master on the right
5	Your first look	Pick Laser Lines on L1, push opacity up	A layer at 0% is invisible; watch PROGRAM change
6	Clips and looks	Clips tab, drop a file on a slot; Looks tab	Banks, auto-play, 67 looks, overlays
7	FX and the beat	Filters tab, Master FX Kaleido then OFF; TAP; LINK	Two FX per layer plus master; strobe locked to the beat and capped at 4/s
8	Scenes	Store a scene, recall it with a fade	128 scenes; the APC and console can recall them
9	The APC mini	The /apc page beside the real APC; walk the modes	PAD MODE changes every row; SHIFT combos; stacks and bumps for busking
10	Cameras + safety	Hub page, Restart cameras; standby card	Cameras as NDI layers; standby between sets; strobe cap

Shooting tips

- Record the screen at 1280 px wide or more, so the PREVIEW and PROGRAM monitors show.
- Use the **Slate** theme on screen; switch to **Stage** once to show it off.
- Turn on **? → Helpers** for the tour chapter: the hover tips explain each control for you.
- Keep a camera on the APC mini for chapter 9, beside the screen recording.
- Before shooting, save your look (store a scene); after, recall it.

24 Glossary

WORD	MEANING
Layer	One slice of the picture stack: BG, L1, L2, L3, L4
BG	The bottom layer, usually the fractal
Opacity	How visible a layer is, 0 to 100%
Blend	How a layer mixes with the layers under it
Source	What a layer shows: clip, fractal, generator or NDI
Generator (GEN)	A pattern drawn live: lines, rays, tunnel and so on
Look	A saved generator setting you fire in one click
Overlay	A second pattern drawn on top of a layer's content
Stack	A whole multi-layer look on one APC pad
Bump	A hold-to-act move that snaps back on release
FX hit	A master effect that flashes while you hold the pad
Scene	A saved full look that recalls with a fade
Master FX	One effect on the whole output
Beat mod	A setting that moves in time with the tempo
Ableton Link	Network tempo sync between music apps and the player
Program	The live output the audience sees
Preview	One layer on its own, for checking before it goes live
Previs	A copy of the output for planning on another screen (sent over NDI)
NDI	Video over the network, used here for previs and cameras
sACN / DMX	How the grandMA3 sends its control values (universe 5)
OSC	Network commands to the player, port 7000
GDTF	The fixture file that tells the grandMA3 the player's channels
Standby card	The logo screen shown at start and between sets
Test card	A pattern for lining up the LED wall or projector

VJ VIZZIES

Scan for the latest download

oats-apps-hub.vercel.app/vizzies